Topic 6: Solved Problems

Step-by-step solutions of all 10 problems of the electrical part of the official sheet (Problems T6andT7_Electric and Magnetic Properties, problems 1 to 10: resistivity and temperature, conductivity and strengthening, ionic and metallic conductors, intrinsic and extrinsic semiconductors, from Session 13). Problems 11 and 12 of the sheet are magnetic and belong to Topic 7. The sheet has a numeric key for some items and none for the reasoning questions, so every solution is derived here and compared with that key where it exists.

`Problem 1` `Problems T6andT7_Electric and Magnetic Properties.pdf #1` `Resistivity · temperature coefficient`

Silver and Aluminium Wires with the Same Resistivity

`Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf`

1. At what temperature will a silver wire exhibit the same resistivity as an aluminium wire at 45 ºC. The resistivities of silver and aluminium at 0 ºC are $1.47 \cdot 10^{-6}$ *and* $2.70 \cdot 10^{-6}$ *Ω·cm respectively and their resistivity coefficients, 0.0038 and 0.0039 ºC⁻¹, respectively.*

`Official Solution: 305 °C`

`HYPOTHESES & PARAMETERS`

Hypotheses

1. **Matthiessen's rule.** The resistivity of a metal is the sum of a thermal part and a residual part, [Session 13 Slide 14]. The residual part (impurities and defects) is already contained in the resistivity measured at 0 °C, so each wire is described by one number ρ_0 and one coefficient α_T.
2. **Linear thermal resistivity.** The thermal part grows linearly with the temperature measured from 0 °C, with the temperature coefficient of resistivity α_T [Session 13 Slide 14]. The sheet gives one constant α_T per metal, so it is used over the whole range up to the result (an extrapolation that the sheet asks for; real silver departs slowly from a straight line).
3. **Same wire geometry is irrelevant.** Resistivity is a property of the material, not of the length or section, so the wires need no dimensions.

Data

Quantity	Symbol	Value as printed	In SI
Resistivity of Ag at 0 °C	$\rho_{0,\mathrm{Ag}}$	1.47×10^{-6} Ω·cm	1.47×10^{-8} Ω·m
Resistivity of Al at 0 °C	$\rho_{0,\mathrm{Al}}$	2.70×10^{-6} Ω·cm	2.70×10^{-8} Ω·m
Coefficient of Ag	α_{Ag}	0.0038 °C⁻¹	0.0038 K⁻¹
Coefficient of Al	α_{Al}	0.0039 °C⁻¹	0.0039 K⁻¹

Quantity	Symbol	Value as printed	In SI
Temperature of the Al wire	T_{Al}	45 °C	45 °C (a difference from 0 °C: 45 K)

Unknown: the temperature T_{Ag} at which silver has the resistivity of the aluminium wire at 45 °C.

The sheet prints the resistivities in Ω·cm and without the Ω and the dot in the first value; they are read from Slide 15, where the same problem is solved (ρ_0 in $\mu\Omega$·cm on Slide 14: Al 2.7, Ag 1.47).

MODEL & STRATEGY

Aluminium is more resistive than silver at 0 °C, so silver has to be heated until the growth of its thermal resistivity makes up the difference. The figure shows the two straight lines and the horizontal level that fixes the answer.

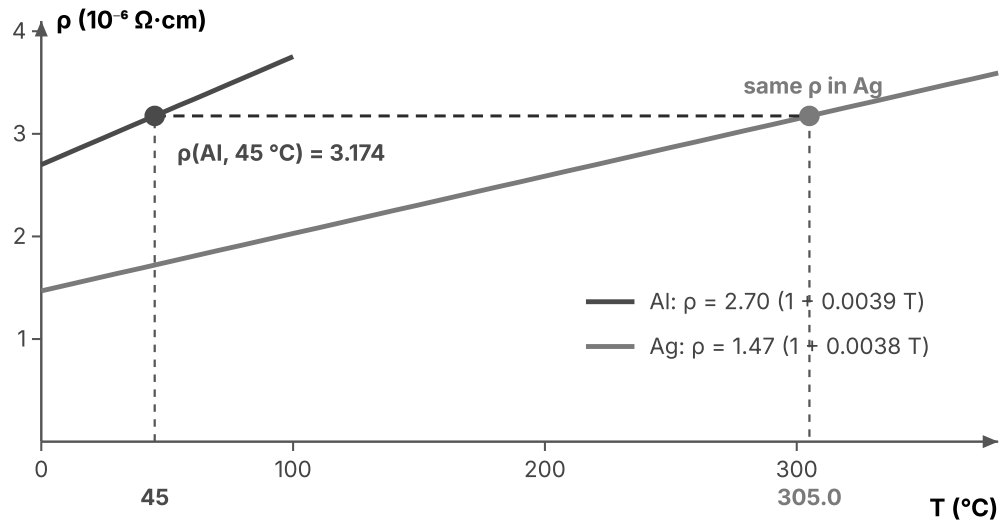


Figure 1. Problem 1: ρ = ρ0 (1 + αT) for Al and Ag; the Al wire at 45 °C has ρ = 3.174×10⁻⁶ Ω·cm and silver reaches it at 305.0 °C.

Strategy.

1. Convert the resistivities to SI.
2. Evaluate the resistivity of aluminium at 45 °C with the law of Slide 14.
3. Impose the same resistivity on silver and solve the linear equation for its temperature.
4. Check by substituting back, and compare with the solution that Slide 15 prints.

STEP-BY-STEP CALCULATION

1. Law of the thermal resistivity [Session 13 Slide 14]

The resistivity at temperature T (in °C, from 0 °C) of each wire is:

$$\rho_T = \rho_0 \left(1 + \alpha_T \, T\right)$$

2. Unit conversion to SI

One ohm-centimetre is one ohm times one hundredth of a metre:

$$1\ \Omega\,\mathrm{cm} = 1\ \Omega \times 10^{-2}\ \mathrm{m} = 10^{-2}\ \Omega\,\mathrm{m}$$

Applied to each resistivity at 0 °C:

$$\rho_{0,\mathrm{Al}} = 2.70 \times 10^{-6}\ \Omega\,\mathrm{cm} \times 10^{-2}\ \frac{\Omega\,\mathrm{m}}{\Omega\,\mathrm{cm}} = 2.70 \times 10^{-8}\ \Omega\,\mathrm{m}$$

$$\rho_{0,\mathrm{Ag}} = 1.47 \times 10^{-6}\ \Omega\,\mathrm{cm} \times 10^{-2}\ \frac{\Omega\,\mathrm{m}}{\Omega\,\mathrm{cm}} = 1.47 \times 10^{-8}\ \Omega\,\mathrm{m}$$

3. Resistivity of the aluminium wire at 45 °C

The product αT is dimensionless:

$$\alpha_{\mathrm{Al}}\, T_{\mathrm{Al}} = 0.0039\ {}^{\circ}\mathrm{C}^{-1} \times 45\ {}^{\circ}\mathrm{C} = 0.1755$$

$$1 + \alpha_{\mathrm{Al}} T_{\mathrm{Al}} = 1 + 0.1755 = 1.1755$$

$$\rho_{\mathrm{Al}}(45\ ^\circ\mathrm{C}) = 2.70 \times 10^{-8}\ \Omega\,\mathrm{m} \times 1.1755 = \mathbf{3.17385 \times 10^{-8}\ \Omega\,m}$$

4. Condition of the statement: the same resistivity in silver

$$\rho_{0,\mathrm{Ag}}\left(1 + \alpha_{\mathrm{Ag}}\,T_{\mathrm{Ag}}\right) = \rho_{\mathrm{Al}}(45\ ^\circ\mathrm{C})$$

Dividing both sides by $\rho_{0,\mathrm{Ag}}$, subtracting 1 and dividing by α_{Ag}:

$$\begin{aligned} 1 + \alpha_{\mathrm{Ag}} T_{\mathrm{Ag}} &= \frac{\rho_{\mathrm{Al}}(45\ ^\circ\mathrm{C})}{\rho_{0,\mathrm{Ag}}} \\ \alpha_{\mathrm{Ag}} T_{\mathrm{Ag}} &= \frac{\rho_{\mathrm{Al}}(45\ ^\circ\mathrm{C})}{\rho_{0,\mathrm{Ag}}} - 1 = \frac{\rho_{\mathrm{Al}}(45\ ^\circ\mathrm{C}) - \rho_{0,\mathrm{Ag}}}{\rho_{0,\mathrm{Ag}}} \\ T_{\mathrm{Ag}} &= \frac{\rho_{\mathrm{Al}}(45\ ^\circ\mathrm{C}) - \rho_{0,\mathrm{Ag}}}{\rho_{0,\mathrm{Ag}}\,\alpha_{\mathrm{Ag}}} \end{aligned}$$

5. Numerical evaluation, one operation per block

Numerator, in ohm-metres:

$$\rho_{\mathrm{Al}}(45\ ^\circ\mathrm{C}) - \rho_{0,\mathrm{Ag}} = 3.17385 \times 10^{-8}\ \Omega\,\mathrm{m} - 1.47 \times 10^{-8}\ \Omega\,\mathrm{m} = 1.70385 \times 10^{-8}\ \Omega\,\mathrm{m}$$

Denominator, in ohm-metres per degree:

$$\rho_{0,\mathrm{Ag}}\,\alpha_{\mathrm{Ag}} = 1.47 \times 10^{-8}\ \Omega\,\mathrm{m} \times 0.0038\ ^\circ\mathrm{C}^{-1} = 5.586 \times 10^{-11}\ \frac{\Omega\,\mathrm{m}}{^\circ\mathrm{C}}$$

Quotient:

$$T_{\mathrm{Ag}} = \frac{1.70385 \times 10^{-8}\ \Omega\,\mathrm{m}}{5.586 \times 10^{-11}\ \Omega\,\mathrm{m}\,^\circ\mathrm{C}^{-1}} = \mathbf{305.0\ ^\circ C}$$

6. Substitution back into the law

$$\alpha_{\mathrm{Ag}} T_{\mathrm{Ag}} = 0.0038\ ^{\circ}\mathrm{C}^{-1} \times 305.02\ ^{\circ}\mathrm{C} = 1.15908$$

$$\rho_{\mathrm{Ag}}(305.02\ ^{\circ}\mathrm{C}) = 1.47 \times 10^{-8}\ \Omega\,\mathrm{m} \times (1 + 1.15908) = 1.47 \times 10^{-8}\ \Omega\,\mathrm{m} \times 2.15908 = 3.17385 \times 10^{-8}\ \Omega\,\mathrm{m}$$

7. The solution printed on Slide 15

Slide 15 rounds the aluminium resistivity to 3.173×10^{-6} Ω·cm before dividing:

$$T_{\mathrm{Ag}} = \frac{3.173 \times 10^{-6} - 1.47 \times 10^{-6}}{1.47 \times 10^{-6} \times 0.0038}\ ^{\circ}\mathrm{C} = \frac{1.703 \times 10^{-6}}{5.586 \times 10^{-9}}\ ^{\circ}\mathrm{C} = 304.87\ ^{\circ}\mathrm{C}$$

RESULT & CHECKS

Checks

- **Units of the coefficient term.**

$$[\alpha_T\, T] = {}^{\circ}\mathrm{C}^{-1} \cdot {}^{\circ}\mathrm{C} = 1$$

- **Units of the result.**

$$[T_{\mathrm{Ag}}] = \frac{\Omega\,\mathrm{m}}{\Omega\,\mathrm{m} \cdot {}^{\circ}\mathrm{C}^{-1}} = {}^{\circ}\mathrm{C}$$

- **Substitution.** Silver at 305.02 °C has 3.17385×10^{-8} Ω·m, equal to aluminium at 45 °C (step 6).
- **Rounding.** The 304.87 of Slide 15 differs from our 305.02 only because the slide rounds ρ_{Al} to four figures; our value keeps the exact product.

Interpretation

Silver conducts better than aluminium at the same temperature, so the temperature that equalises them is far above 45 °C: silver must be heated until its thermal resistivity has grown from its low value at 0 °C up to the aluminium value (step 5). The result is large because the thermal slope of

silver, $\rho_{0,\mathrm{Ag}}\alpha_{T,\mathrm{Ag}}$, is small; as that slope tends to zero the temperature tends to infinity. The linear law is used up to 305 °C because the sheet fixes it; real metals bend away from a straight line at high temperature.

Check against the official solution

Item	Official	Computed	Comment
Temperature of the silver wire	305 °C	305.0 °C	agrees; Slide 15 prints 304.86 because it rounds ρ_{Al} to 3.173 (we get 304.87 with that rounding)

Problem 2	Problems T6andT7_Electric and Magnetic Properties.pdf #2	Conductivity · strengthening · Matthiessen

Strategies to Strengthen an Aluminium Wire and Their Effect on its Conductivity

Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf

2. Reason what effect the following strategies would have on the conductivity and tensile strength of an aluminum alloy wire:

- *a) Alloying with a more conductive metal as copper.*
- *b) Decrease on its working temperature.*
- *c) Thermal treatment that increases the grain size of the alloy.*
- *d) Treatment that causes intermetallic precipitation.*
- *e) Metal drawing (forming) that involves a high % of cold work.*

Indicate which would be the best strategy to reduce the weight of the wire while maintaining its tensile strength and conductivity.

Official Solution: a) conductivity: decrease; strength: increase; b) conductivity: increase; strength: more increase; c) conductivity: increase; strength: decrease; d) conductivity: decrease; strength: increase; e) conductivity: decrease; strength: increase

HYPOTHESES & PARAMETERS

Hypotheses

1. **Matthiessen's rule.** The total resistivity of a metal is the thermal part plus the residual part, and the residual part is the sum of an impurity part and a deformation part [Session 13 Slides 14, 16]. Conductivity is the inverse of resistivity [Session 13 Slide 13].
2. **Scattering lowers conductivity.** Anything that distorts the metal lattice scatters the conduction electrons, reduces their mobility and therefore the conductivity [Session 13 Slides 16-17].
3. **All hardening methods lower the conductivity** to a greater or lesser extent [Session 13 Slide 18], and they raise the strength by making dislocation glide harder (Topic 5).
4. **Notation.** The sheet uses "σ" for conductivity in this topic and for stress in Topic 5; here the electrical conductivity is γ and the tensile strength is σ_{ts}, to keep both in the same equation without confusion.
5. **Same alloy density and length.** For the weight comparison the density and the length of the wire do not change with the strategy; only the section A does.

Data

Quantity	Symbol	Meaning
Electrical conductivity	$\gamma = 1/\rho$	property to keep or improve
Tensile strength	σ_{ts}	property to keep or improve
Wire section	A	unknown to reduce (mass $\propto A$)

Unknown: the direction of change of γ and σ_{ts} for each strategy, and the strategy that allows the lightest wire.

MODEL & STRATEGY

Each strategy acts on one term of Matthiessen's rule and, at the same time, on dislocation motion. The figure collects the directions of the key: only strategy b moves into the quadrant in which both properties improve.

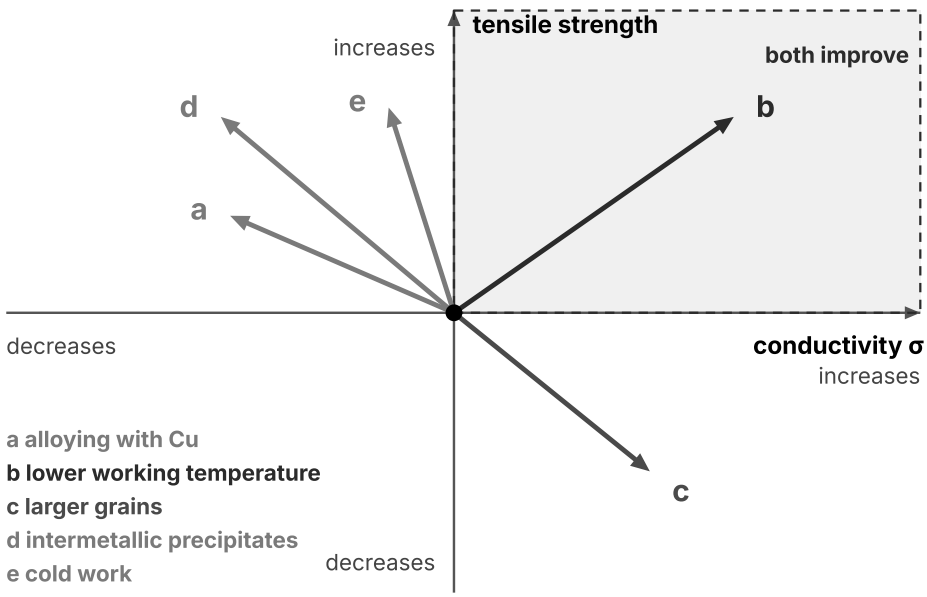


Figure 2. Problem 2: direction of the change of conductivity (horizontal) and tensile strength (vertical) for the five strategies a to e (arrows are schematic, not to scale).

Strategy.

1. Write Matthiessen's rule and decide, for each strategy, which resistivity term changes and in which direction.
2. Read the effect on the strength from Topic 5 (strengthening mechanisms) and from the chart of Slide 18.
3. For the weight question, write the three conditions of a wire of section A' (load capacity, conductance, mass) and find when $A' < A$ is allowed.

STEP-BY-STEP CALCULATION

1. The law that governs the conductivity [Session 13 Slides 13, 14, 16]

Total resistivity as the sum of the thermal, impurity and deformation parts, and its inverse:

$$\rho_{\text{total}} = \rho_T + \rho_r = \rho_T + \rho_i + \rho_d$$

$$\gamma = \frac{1}{\rho_{\text{total}}} = \frac{1}{\rho_T + \rho_i + \rho_d}$$

A larger sum in the denominator means a smaller conductivity; a smaller sum means a larger one.

2. Strategy a: alloying with copper

The solute atoms are foreign atoms in the aluminium lattice and scatter the electrons whatever their own conductivity (Slide 16 shows that every addition raises the resistivity of copper); copper dissolved in aluminium therefore adds to ρ_i (Topic 5: solid solution strengthening):

$$\rho_i \uparrow \Rightarrow\ \rho_{\text{total}} \uparrow \Rightarrow\ \gamma = \frac{1}{\rho_{\text{total}}} \downarrow \qquad \sigma_{ts} \uparrow$$

3. Strategy b: lower working temperature

The thermal term is $\rho_T = \rho_0(1 + \alpha_T T)$ [Session 13 Slide 14], which falls when T falls:

$$T \downarrow \Rightarrow\ \rho_T \downarrow \Rightarrow\ \rho_{\text{total}} \downarrow \Rightarrow\ \gamma \uparrow$$

The key also gives a strength increase, larger than the others. This is the reading of the key, not a derivation from the slides; the usual reason is that at lower temperature the atoms vibrate less and thermally assisted dislocation motion is weaker, so a higher stress is needed to move dislocations (a gloss, not content of Session 13):

$$T \downarrow \Rightarrow\ \sigma_{ts} \uparrow\uparrow$$

4. Strategy c: grain growth

A larger grain has fewer grain boundaries per unit volume, and the boundaries are part of the deformation term [Session 13 Slide 16]; at the same time the Hall-Petch effect of Topic 5 (grain size strengthening) weakens the metal:

$$d_{\text{grain}} \uparrow \Rightarrow \text{ boundaries} \downarrow \Rightarrow \rho_d \downarrow \Rightarrow \gamma \uparrow \qquad \sigma_{ts} \downarrow$$

5. Strategy d: intermetallic precipitation

The precipitates are second-phase particles that scatter electrons and block dislocations (Topic 5: precipitation strengthening), and the matrix keeps a part of the solute that raises its impurity term [Session 13 Slide 16]:

$$\text{precipitates} \Rightarrow \rho_{\text{total}} \uparrow \Rightarrow \gamma \downarrow \qquad \sigma_{ts} \uparrow$$

6. Strategy e: heavy cold work

Drawing multiplies dislocations, which are structural imperfections of the deformation term [Session 13 Slide 16] (Topic 5: strain hardening). The chart of Slide 18 shows the yield strength rising strongly with the cold work while the conductivity falls only slightly:

$$\text{cold work} \uparrow \Rightarrow \rho_d \uparrow \Rightarrow \gamma \downarrow \text{ (slightly)} \qquad \sigma_{ts} \uparrow$$

7. The lightest wire that keeps strength and conductivity

Load capacity, conductance and mass of the original wire, of length l and density ρ_m, and of the new wire of section A' with properties γ' and σ'_{ts}:

$$F_{\max} = \sigma_{ts} A \qquad G = \frac{\gamma A}{l} \qquad m = \rho_m A l$$

$$F'_{\max} = \sigma'_{ts} A' \qquad G' = \frac{\gamma' A'}{l} \qquad m' = \rho_m A' l$$

The new wire must carry at least the same load and conduct at least as well:

$$\sigma'_{ts}\,A' \geq \sigma_{ts}\,A \;\Rightarrow\; \frac{A'}{A} \geq \frac{\sigma_{ts}}{\sigma'_{ts}}$$

$$\frac{\gamma'\,A'}{l} \geq \frac{\gamma\,A}{l} \;\Rightarrow\; \frac{A'}{A} \geq \frac{\gamma}{\gamma'}$$

Both conditions together give a lower bound for the section, and the mass ratio equals the section ratio because ρ_m and l are unchanged:

$$\frac{m'}{m} = \frac{A'}{A} \geq \max\left(\frac{\sigma_{ts}}{\sigma'_{ts}},\; \frac{\gamma}{\gamma'}\right)$$

A lighter wire means $m'/m < 1$, which is possible only if the larger of the two ratios on the right is below 1, that is, if both $\sigma'_{ts} > \sigma_{ts}$ and $\gamma' > \gamma$:

$$m' < m \iff \sigma'_{ts} > \sigma_{ts} \text{ and } \gamma' > \gamma$$

Applied to the five strategies of the key:

Strategy	γ'/γ	σ'_{ts}/σ_{ts}	**Lighter wire possible?**
a alloying with Cu	< 1	> 1	no: conductance fixes $A'/A \geq \gamma/\gamma' > 1$
b lower temperature	> 1	> 1	yes
c grain growth	> 1	< 1	no: load capacity fixes $A'/A \geq \sigma_{ts}/\sigma'_{ts} > 1$
d precipitates	< 1	> 1	no: conductance
e cold work	< 1	> 1	no: conductance

RESULT & CHECKS

Checks

- **Key.** The directions of a to e reproduce the key of the sheet term by term.
- **Units of the weight argument.** Each ratio is dimensionless, and the three conditions are homogeneous:

$$[F_{\max}] = \mathrm{Pa}\cdot\mathrm{m}^2 = \mathrm{N} \qquad [G] = \frac{\Omega^{-1}\mathrm{m}^{-1}\cdot\mathrm{m}^2}{\mathrm{m}} = \Omega^{-1} \qquad [m] = \frac{\mathrm{kg}}{\mathrm{m}^3}\cdot\mathrm{m}^2\cdot\mathrm{m} = \mathrm{kg}$$

- **Limit.** If $\gamma' = \gamma$ and $\sigma'_{ts} = \sigma_{ts}$ the bound is $A'/A \geq 1$: no strategy, no saving.

Interpretation

Strategies a, d and e buy strength with conductivity and strategy c does the opposite, so none of them can shrink the wire when both properties are required. Only a lower working temperature improves both, so it is the strategy that permits a lighter wire. This choice of "best" is the reasoning of this note from the conditions above; the key of the sheet lists only the directions of change. A lower working temperature needs cooling in service, which has its own cost and mass; a material change is the alternative route.

Check against the official solution

Item	Official	Computed	Comment
a) alloying with Cu	γ decreases; strength increases	γ decreases; strength increases	agrees
b) lower temperature	γ increases; strength more increase	γ increases; strength increases most	agrees
c) larger grains	γ increases; strength decreases	γ increases; strength decreases	agrees
d) precipitates	γ decreases; strength increases	γ decreases; strength increases	agrees
e) cold work	γ decreases; strength increases	γ decreases; strength increases	agrees
Best strategy for a lighter wire	not printed	b	reasoning of this note (both properties must rise)

Problem 3 | Problems T6andT7_Electric and Magnetic Properties.pdf #3 | Ionic and metallic conduction · defects

Ionic Ceramic against Metal: Defects, Temperature and Doping

Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf

3. Compare the effect of following factors on the conductivity of an ionic conductor and a metallic conductor

Factor	Ionic ceramic	Metal
Vacant positions in the net		
Cations in interstitial positions		
Anions in interstitial positions		
Increase of temperature		
Disorder of the net		
Doping with substances with other valences		

The sheet prints no solution for this problem.

HYPOTHESES & PARAMETERS

Hypotheses

1. **Metal.** The conduction electrons are free and their number is fixed by the atoms; the conductivity is $\sigma = n|q_e|\mu$, and anything that scatters electrons lowers the mobility μ [Session 13 Slides 12-13]. Resistivity is the sum of the thermal, impurity and deformation terms [Session 13 Slides 14, 16].
2. **Ionic ceramic.** The charge is carried by ions: cations or anions that move through vacancies or interstitial voids, at high temperature [Session 13 Slide 22]. The number of mobile ions and their ease of movement decide the conductivity, so the same structure formula $\sigma = \sum n_k|q_k|\mu_k$ holds with ions in place of electrons (an extension of the form of Slide 12, which the slide writes for electrons only).
3. **Cations are the small ions.** The slide draws the interstitial movement for cations of radius smaller than the atom ($r_i < r_{atom}$) [Session 13 Slide 22]; the anions of an ionic crystal are the larger ions and fit interstitial voids poorly (Topic 2).
4. **The ionic column is argued** from Slide 22 and from the defect chemistry of Topic 2; the metal column from Slides 13, 14 and 16. The sheet prints no key, so no verdict below is copied from a solution.

Data

Six factors, two kinds of conductor; the unknown is the sign of the effect on the conductivity (↑ increases, ↓ decreases, ≈ almost no effect).

MODEL & STRATEGY

The two columns follow opposite logic: in the metal a defect is an obstacle for the electron, in the ionic ceramic a defect is a place where an ion can jump. The figure shows both pictures.

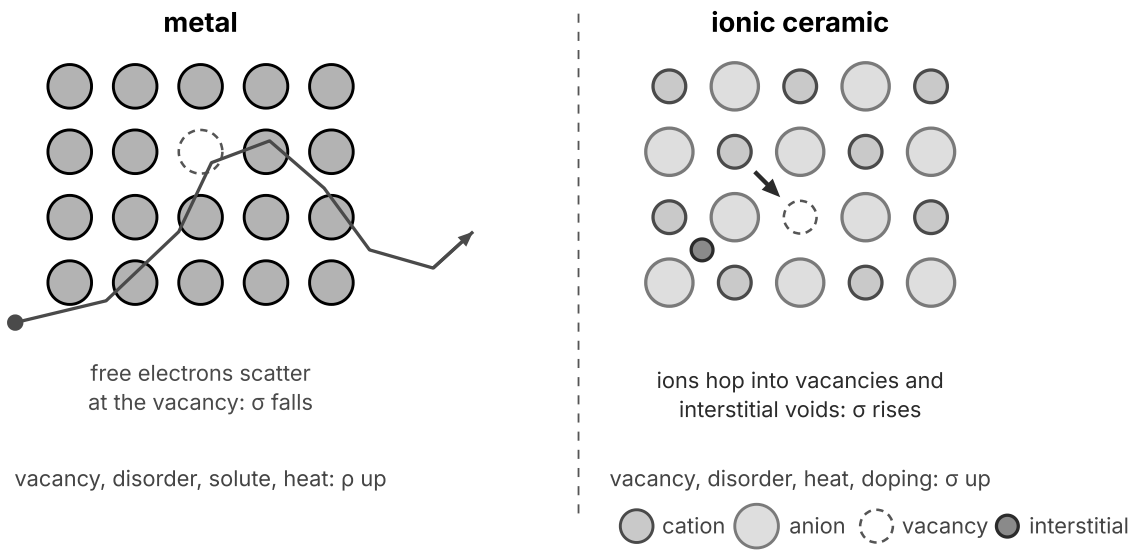


Figure 3. Problem 3: in a metal the electrons scatter at a vacancy; in an ionic ceramic the cations hop into vacancies and move through interstitial voids (schematic).

Strategy.

1. Write the conductivity of each conductor type as number of carriers times charge times mobility.
2. For each factor decide whether it changes the number of mobile carriers or their mobility.
3. Collect the verdicts in a table.

STEP-BY-STEP CALCULATION

1. Metallic conductor [Session 13 Slides 12-14, 16]

Conductivity from the number of free electrons and their mobility, and the mobility from the relaxation time between collisions:

$$\sigma = n\,|q_e|\,\mu \qquad \mu = \frac{\tau\,|q_e|}{m_e}$$

Every factor that increases the number of collisions shortens τ, lowers μ and so lowers σ; Matthiessen's rule collects the sources:

$$\rho_{\text{total}} = \rho_T + \rho_i + \rho_d \qquad \sigma = \frac{1}{\rho_{\text{total}}}$$

The thermal term grows with temperature:

$$\rho_T = \rho_0 \left(1 + \alpha_T \, T\right)$$

2. Ionic conductor [Session 13 Slide 22]

Conductivity as the sum over the mobile ionic species, with the number of mobile ions n_k and their mobility μ_k (extension of the form of Slide 12):

$$\sigma_{\text{ionic}} = \sum_k n_k \left|q_k\right| \mu_k$$

The ions move by jumps between lattice sites, so their mobility follows the diffusivity of the ion, which is thermally activated (Topic 3, Arrhenius):

$$D = D_0 \exp\!\left(-\frac{Q}{R\,T}\right)$$

and the Nernst-Einstein relation links mobility and diffusivity (extension, not in Session 13):

$$\mu_k = \frac{\left|q_k\right| D_k}{k_B \, T}$$

Jumps need empty sites. The equilibrium fraction of vacancies grows with temperature (Topic 2, thermal vacancies):

$$\frac{n_v}{N} = \exp\!\left(-\frac{Q_v}{k_B \, T}\right)$$

3. Vacant positions and interstitial atoms

In the metal a vacancy or an interstitial atom distorts the lattice and belongs to the deformation or impurity term:

$$n_v \uparrow \Rightarrow \ \rho_d \uparrow \Rightarrow \ \sigma \downarrow$$

In the ionic crystal a vacancy offers the neighbouring cations an empty site to jump into, and an interstitial cation can move through the interstitial voids [Session 13 Slide 22]:

$$n_v \uparrow \Rightarrow \ \text{jumps} \uparrow \Rightarrow \ \mu_k \uparrow \Rightarrow \ \sigma_{\text{ionic}} \uparrow$$

An interstitial anion is the exception: the anion is the larger ion and fits an interstitial void only with a large distortion, so few anions are interstitial and they hardly move; the slide draws the interstitial jump for the small cation. The verdict is "almost no effect" (this is a size argument of Topic 2, not a statement of the slide):

$$r_{\text{anion}} > r_{\text{cation}} \ \Rightarrow \ \text{few mobile interstitial anions} \ \Rightarrow \ \Delta\sigma_{\text{ionic}} \approx 0$$

4. Temperature

Metal: the thermal resistivity grows linearly and the conductivity falls:

$$T \uparrow \Rightarrow \ \rho_T = \rho_0(1 + \alpha_T T) \uparrow \Rightarrow \ \sigma \downarrow$$

Ionic ceramic: both the number of vacancies and the diffusivity grow exponentially with T, so the conductivity rises strongly (conduction occurs "at high temperatures" [Session 13 Slide 22]):

$$T \uparrow \Rightarrow \ \exp\left(-\frac{Q_v}{k_B T}\right) \uparrow \ \text{and} \ \exp\left(-\frac{Q}{RT}\right) \uparrow \Rightarrow \ \sigma_{\text{ionic}} \uparrow\uparrow$$

5. Disorder of the net

Metal: disorder is the distortion of the metal net; the residual resistivity increases with "the degree of distortion" [Session 13 Slide 16]:

$$\text{disorder} \uparrow \Rightarrow \ \rho_r \uparrow \Rightarrow \ \sigma \downarrow$$

Ionic ceramic: disorder (point defects, open structure) means more empty sites and more paths for the jumps:

$$\text{disorder} \uparrow \Rightarrow \ \text{empty sites and paths} \uparrow \Rightarrow \ \sigma_{\text{ionic}} \uparrow$$

6. Doping with substances of another valence

Metal: a solute is an impurity, whatever its valence [Session 13 Slide 16]:

$$\text{solute} \ \Rightarrow \ \rho_i \uparrow \Rightarrow \ \sigma \downarrow$$

Ionic ceramic: an ion of another valence must be compensated to keep the crystal neutral. Two Al^{3+} replacing two Mg^{2+} in MgO bring an excess charge that a cation vacancy removes (the case of Problem T2-DEF03):

$$2 \times (+3) - 2 \times (+2) = +2 \ \Rightarrow \ \text{one cation vacancy of charge } -2$$

so doping creates vacancies and the conductivity rises:

$$\text{aliovalent doping} \ \Rightarrow \ n_v \uparrow \Rightarrow \ \sigma_{\text{ionic}} \uparrow$$

7. Result table

Factor	Ionic ceramic	Metal
Vacant positions in the net	↑	↓
Cations in interstitial positions	↑	↓
Anions in interstitial positions	≈	↓
Increase of temperature	↑↑	↓
Disorder of the net	↑	↓

Factor	Ionic ceramic	Metal
Doping with substances with other valences	↑	↓

RESULT & CHECKS

Checks

- **Units of the ionic expression.**

$$[\sigma_{\text{ionic}}] = \mathrm{m}^{-3} \cdot \mathrm{C} \cdot \frac{\mathrm{m}^2}{\mathrm{V\,s}} = \frac{\mathrm{A}}{\mathrm{V\,m}} = \Omega^{-1}\mathrm{m}^{-1}$$

- **Opposite logic.** Every entry of the metal column is a decrease; five of six entries of the ionic column are an increase. In the metal the carriers (electrons) are not created by the defects, so a defect can only hinder them; in the ionic crystal the defects are both the carriers' route and their number.
- **Limits.** At a very low temperature the vacancies freeze out and an ionic ceramic is an insulator, while a metal has its lowest resistivity (only ρ_r left); at a very high temperature the ceramic is a good ionic conductor and the metal is at its most resistive.

Interpretation

The same defect has opposite meaning in the two conductors: it is an obstacle for an electron wave in the metal and a stepping stone for an ion in the ceramic. This is why metals are cooled to improve conduction and ceramic electrolytes (fuel cells, gas sensors, Slide 22) are heated to work.

Check against the official solution

Item	Official	Computed	Comment
All six factors	not printed on the sheet	table of step 7	argued from Session 13 Slides 13, 14, 16, 22 and Topics 2-3; the interstitial anion entry and the Nernst-Einstein relation are the note's own reasoning, not slide content

Problem 4 | Problems T6andT7_Electric and Magnetic Properties.pdf #4 | Metal vs intrinsic semiconductor · temperature

Effect of Temperature on a Metal and on an Intrinsic Semiconductor

Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf, p. 2

> *4. Justify the different effects that an increase on the exposure temperature have on a metal and on an intrinsic semiconductor*
>
> *The sheet prints no solution for this problem.*

HYPOTHESES & PARAMETERS

Hypotheses

1. **Conductivity is carriers times mobility.** For any conductor $\sigma = n|q|\mu$; for an intrinsic semiconductor both electrons and holes count, $\sigma = n|q|(\mu_e + \mu_h)$ [Session 13 Slides 12, 27-28].
2. **Metal.** The conduction electrons are the non-bonded electrons of the atoms, so their number does not depend on temperature in the range considered; only the mobility changes, through the thermal resistivity $\rho_T = \rho_0(1 + \alpha_T T)$ [Session 13 Slides 13-14].
3. **Intrinsic semiconductor.** The carriers are created by thermal excitation across the gap, so $n = p = n_i$ grows with the temperature as $n_i \propto \exp(-E_g/2kT)$ and the exponential factor is the one that dominates the conductivity [Session 13 Slide 29]. The temperature also disorders the lattice, but that effect "is masked by the increase of the number of charge carriers" (Slide 29).
4. **Numerical illustration.** Copper with $\alpha_T = 0.0039$ °C⁻¹ [Session 13 Slide 14] and germanium with $E_g = 0.67$ eV [Session 13 Slide 8], both from 25 °C to 75 °C, the interval of Problem T6-07. $k_B = 8.62 \times 10^{-5}$ eV/K.

Data

Quantity	Symbol	Value
Temperature coefficient of Cu	α_T	0.0039 °C⁻¹
Band gap of Ge	E_g	0.67 eV
Boltzmann constant	k_B	8.62×10^{-5} eV/K
Temperatures	T_1, T_2	25 °C, 75 °C (298.15 K, 348.15 K)

Unknown: the sign and the size of the change of σ in each material.

MODEL & STRATEGY

Take the logarithmic derivative of $\sigma = n\mu$ with respect to the temperature: it splits the change of the conductivity into the change of the carrier number and the change of the mobility. In the metal only the second term exists; in the intrinsic semiconductor the first term is far larger than the second.

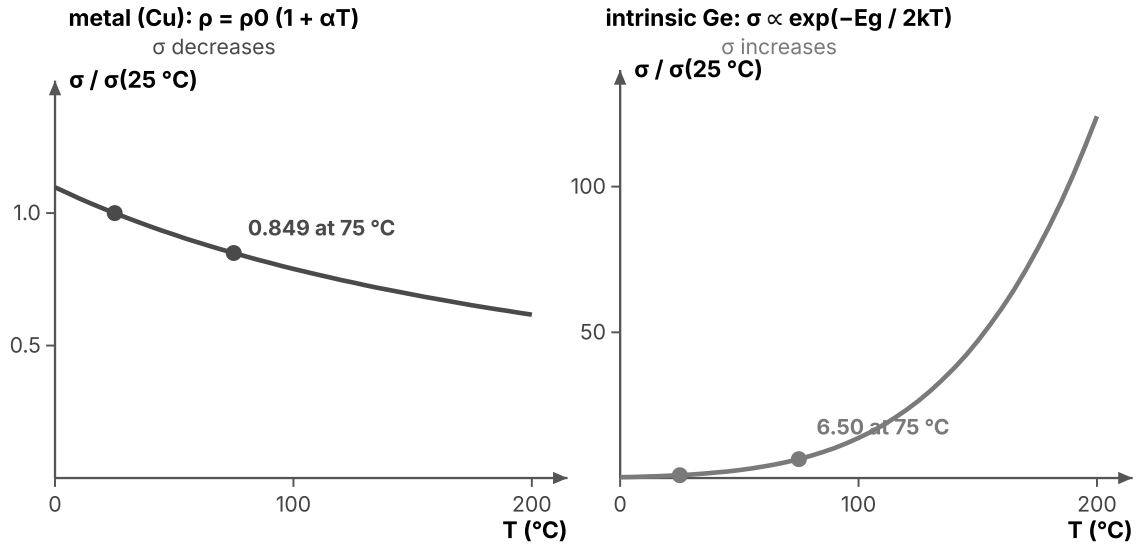


Figure 4. Problem 4: relative conductivity of copper (ρ = ρ0(1 + αT), α = 0.0039 °C^{-1}) and of intrinsic germanium (Eg = 0.67 eV) against temperature, each divided by its value at 25 °C.

Strategy.

1. Write the logarithmic derivative of the conductivity.
2. Evaluate it for the metal, using Slide 14, and for the semiconductor, using Slide 29.
3. Compare the numerical changes between 25 and 75 °C.

STEP-BY-STEP CALCULATION

1. The two ways in which temperature can act

Conductivity as number of carriers times mobility, and its logarithmic derivative (product rule of the logarithm and chain rule):

$$\sigma = n\,|q|\,\mu \;\Rightarrow\; \ln\sigma = \ln n + \ln|q| + \ln\mu$$

$$\frac{1}{\sigma}\frac{d\sigma}{dT} = \frac{1}{n}\frac{dn}{dT} + \frac{1}{\mu}\frac{d\mu}{dT}$$

2. Metal [Session 13 Slides 13-14]

The number of free electrons is constant, so the first term vanishes, and the conductivity is the inverse of the thermal resistivity:

$$\frac{dn}{dT} = 0 \qquad \sigma = \frac{1}{\rho_0 (1 + \alpha_T T)}$$

Derivative of the inverse (chain rule with $u = 1 + \alpha_T T$, $du/dT = \alpha_T$):

$$\frac{1}{\sigma}\frac{d\sigma}{dT} = -\frac{1}{u}\frac{du}{dT} = -\frac{\alpha_T}{1 + \alpha_T T}$$

Evaluated for copper at 25 °C:

$$1 + \alpha_T T = 1 + 0.0039\ {}^\circ\mathrm{C}^{-1} \times 25\ {}^\circ\mathrm{C} = 1.0975$$

$$\frac{1}{\sigma}\frac{d\sigma}{dT} = -\frac{0.0039\ {}^\circ\mathrm{C}^{-1}}{1.0975} = \mathbf{-3.554 \times 10^{-3}\ K^{-1}}$$

so the relative change of the conductivity per kelvin, written as a percentage, is:

$$-3.554 \times 10^{-3}\ \mathrm{K}^{-1} \times 100\ \% = -0.3554\ \%\ \mathrm{K}^{-1}$$

Over the 50 K of the interval:

$$\frac{\rho(75\ {}^\circ\mathrm{C})}{\rho(25\ {}^\circ\mathrm{C})} = \frac{1 + 0.0039 \times 75}{1 + 0.0039 \times 25} = \frac{1.2925}{1.0975} = 1.1777$$

$$\frac{\sigma(75\,^\circ\mathrm{C})}{\sigma(25\,^\circ\mathrm{C})} = \frac{1}{1.1777} = \mathbf{0.8491}$$

3. Intrinsic semiconductor [Session 13 Slide 29]

The conductivity follows the carrier number:

$$\sigma = \sigma_0 \exp\!\left(-\frac{E_g}{2k_B T}\right) \;\Rightarrow\; \ln\sigma = \ln\sigma_0 - \frac{E_g}{2k_B}\,T^{-1}$$

Derivative with respect to T (power rule on T^{-1}):

$$\frac{1}{\sigma}\frac{d\sigma}{dT} = -\frac{E_g}{2k_B}\left(-T^{-2}\right) = \frac{E_g}{2k_B\,T^2}$$

Evaluated for germanium at 298.15 K, first the constant $E_g/2k_B$ and the square of the temperature:

$$\frac{E_g}{2k_B} = \frac{0.67\ \mathrm{eV}}{2 \times 8.62\times10^{-5}\ \mathrm{eV/K}} = 3886.3\ \mathrm{K}$$

$$T^2 = (298.15\ \mathrm{K})^2 = 8.8893\times10^{4}\ \mathrm{K}^2$$

$$\frac{1}{\sigma}\frac{d\sigma}{dT} = \frac{3886.3\ \mathrm{K}}{8.8893\times10^{4}\ \mathrm{K}^2} = +\mathbf{4.372\times10^{-2}\ K^{-1}}$$

so the relative change of the conductivity per kelvin, written as a percentage, is:

$$4.372\times10^{-2}\ \mathrm{K^{-1}} \times 100\ \% = +4.372\ \%\ \mathrm{K^{-1}}$$

Over the 50 K of the interval (the ratio of Problem T6-07):

$$\frac{\sigma(75\ ^\circ\mathrm{C})}{\sigma(25\ ^\circ\mathrm{C})} = \exp\left[3886.3\ \mathrm{K}\left(\frac{1}{298.15\ \mathrm{K}} - \frac{1}{348.15\ \mathrm{K}}\right)\right] = \exp(1.8720) = \mathbf{6.501}$$

4. Comparison of the two

Ratio of the relative rates at 25 °C, in absolute value:

$$\frac{4.372 \times 10^{-2}\ \mathrm{K}^{-1}}{3.554 \times 10^{-3}\ \mathrm{K}^{-1}} = 12.30$$

RESULT & CHECKS

Checks

- **Units of both logarithmic rates.**

$$\left[\frac{\alpha_T}{1+\alpha_T T}\right] = \frac{^\circ\mathrm{C}^{-1}}{1} = \mathrm{K}^{-1} \qquad \left[\frac{E_g}{2k_B T^2}\right] = \frac{\mathrm{eV}}{\mathrm{eV\,K^{-1}\,K^2}} = \mathrm{K}^{-1}$$

- **Interval.** A temperature difference of 1 °C equals 1 K, so the coefficient in °C⁻¹ can be read in K⁻¹.
- **Signs.** Metal: negative rate; semiconductor: positive rate.

Interpretation

In the metal the carriers are already all there; heat only agitates the ions, the electrons collide more often, the mobility falls and so does σ. In the intrinsic semiconductor the heat itself creates carriers by lifting electrons across the gap, and the exponential growth of their number (4.4 % per kelvin) overpowers the loss of mobility, which is the same kind of effect as in the metal (a loss of 0.36 % per kelvin for copper). Over 50 K the metal loses 15 % of its conductivity and the semiconductor multiplies it by 6.5. For $E_g \to 0$ the semiconductor rate tends to zero, as in a metal where the carrier number is temperature independent; for a large gap the rate becomes very large, and the material is an insulator at ordinary temperatures.

Check against the official solution

Item	Official	Computed	Comment
Effect on the metal	not printed on the sheet	σ decreases (0.8491 for Cu, 25 to 75 °C)	consistent with Session 13 Slides 13-14

Item	Official	Computed	Comment
Effect on the intrinsic semiconductor	not printed on the sheet	σ increases (6.501 for Ge, 25 to 75 °C)	consistent with Session 13 Slide 29; the 6.501 is the ratio of Problem T6-07

Problem 5 | Problems T6andT7_Electric and Magnetic Properties.pdf #5 | Extrinsic type · donors and acceptors

Extrinsic Type of Silicon Doped with Sb, As and Ga

Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf, p. 2

5. Indicate which of the following semiconductors are extrinsic n-type and extrinsic p-type and whether the introduced impurities act as donors or acceptors

- *a) Silicon doped with antimony in substitutional solid solution*
- *b) Silicon doped with arsenic in substitutional solid solution*
- *c) Silicon doped with gallium in substitutional solid solution*

Slide 39 poses the same question with germanium doped with arsenic in item b; the classification is the same (step 5).

Official Solution: a) n-type extrinsic; b) n-type extrinsic; c) p-type extrinsic

HYPOTHESES & PARAMETERS

Hypotheses

1. **Substitutional doping.** The dopant atom takes the place of a silicon atom in the diamond lattice, where each atom shares its 4 valence electrons in covalent bonds [Session 13 Slides 26, 31].
2. **Donor and acceptor rule.** An impurity with more valence electrons than the host (group 15) donates the extra electron to the conduction band and is a donor, giving n-type material; an impurity with fewer valence electrons (group 13) leaves a hole and is an acceptor, giving p-type material [Session 13 Slides 35-37].
3. **Group numbers.** Sb and As are in group 15 (VA), Ga is in group 13 (IIIA), Si and Ge are in group 14 (IVA) [Session 13 Slides 35-37]. (Slide 37 prints "group VA" next to B, Al, Ga, but the table on the same slide places them in group 3A.)
4. **Room temperature,** taken as 300 K here (the problems whose statement gives 25 °C use 298.15 K), with $k_B = 8.62 \times 10^{-5}$ eV/K, and the ionisation energies of Slide 40.

Data

Dopant	Group	Valence electrons	Host	Level of Slide 40
Sb	15	5	Si	$E_d = 0.039$ eV
As	15	5	Si	$E_d = 0.049$ eV
Ga	13	3	Si	$E_a = 0.065$ eV
As	15	5	Ge	$E_d = 0.0127$ eV

Unknown: type of each semiconductor (n or p) and role of each impurity (donor or acceptor).

MODEL & STRATEGY

Compare the number of valence electrons of the dopant with the 4 of the host: the difference is the electron surplus (donor) or the hole surplus (acceptor). The figure marks the three dopants of the sheet in the periodic table.

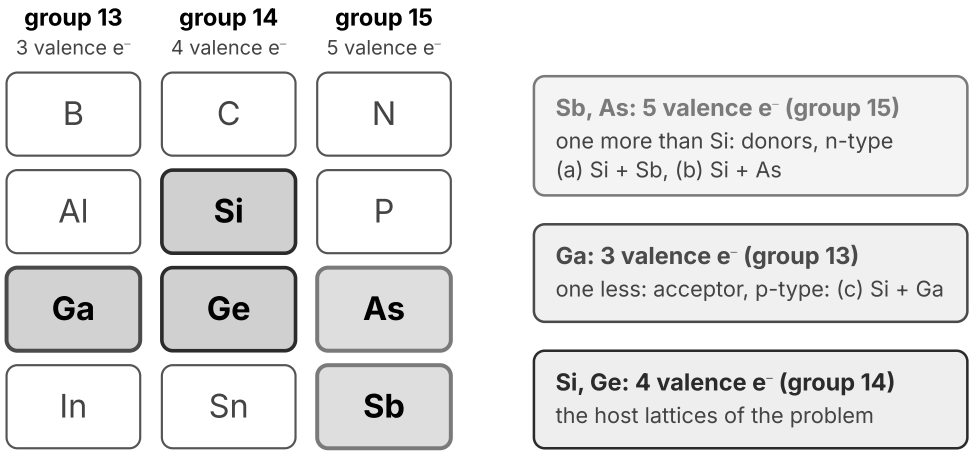


Figure 5. Problem 5: the valence electrons of the dopants (groups 15 and 13) against the 4 of the silicon or germanium host (group 14) decide donor or acceptor behaviour.

Strategy.

1. Count the valence electrons of each dopant and subtract the 4 of the host.
2. Assign donor or acceptor from the sign of the difference, and n or p from the carrier that is in excess.
3. Check with the ionisation energies of Slide 40 that the level lies close enough to its band to be ionised at room temperature.

STEP-BY-STEP CALCULATION

1. Valence-electron surplus of each substitution [Session 13 Slides 36-37]

Surplus of electrons with respect to the 4 valence electrons that the silicon atom replaced:

$$\Delta Z_{\mathrm{Sb}} = Z_{\mathrm{Sb}} - Z_{\mathrm{Si}} = 5 - 4 = +1$$

$$\Delta Z_{\mathrm{As}} = Z_{\mathrm{As}} - Z_{\mathrm{Si}} = 5 - 4 = +1$$

$$\Delta Z_{\mathrm{Ga}} = Z_{\mathrm{Ga}} - Z_{\mathrm{Si}} = 3 - 4 = -1$$

2. Roles and types

Four of the five valence electrons of Sb or As fill the four covalent bonds; the fifth one is loosely bound and jumps to the conduction band from the donor level E_d [Session 13 Slide 36]:

$$\Delta Z = +1 \;\Rightarrow\; \text{donor} \;\Rightarrow\; n_e \gg n_H \;\Rightarrow\; \sigma \approx n_e \, |q_e| \, \mu_e \quad \text{(n-type)}$$

Ga has three valence electrons for four bonds; one bond is incomplete and a neighbouring valence electron can fill it from just above the valence band, leaving a hole [Session 13 Slide 37]:

$$\Delta Z = -1 \;\Rightarrow\; \text{acceptor} \;\Rightarrow\; p \gg n \;\Rightarrow\; \sigma \approx p \, |q_e| \, \mu_h \quad \text{(p-type)}$$

3. Ionisation energies against the thermal energy at 300 K

Thermal energy:

$$k_B T = 8.62 \times 10^{-5} \, \frac{\text{eV}}{\text{K}} \times 300 \text{ K} = 0.02586 \text{ eV}$$

Ratio of each level of Slide 40 to it, with the conversion of the energy to joules ($1 \text{ eV} = 1.602 \times 10^{-19}$ J):

$$E_d(\text{Sb in Si}) = 0.039 \text{ eV} \times 1.602 \times 10^{-19} \, \frac{\text{J}}{\text{eV}} = 6.248 \times 10^{-21} \text{ J} \qquad \frac{0.039}{0.02586} = 1.51$$

$$E_d(\text{As in Si}) = 0.049\ \text{eV} \times 1.602 \times 10^{-19}\ \frac{\text{J}}{\text{eV}} = 7.850 \times 10^{-21}\ \text{J} \qquad \frac{0.049}{0.02586} = 1.89$$

$$E_a(\text{Ga in Si}) = 0.065\ \text{eV} \times 1.602 \times 10^{-19}\ \frac{\text{J}}{\text{eV}} = 1.041 \times 10^{-20}\ \text{J} \qquad \frac{0.065}{0.02586} = 2.51$$

4. Result of the sheet

a) Si + Sb : **n-type**, **Sb donor** b) Si + As : **n-type**, **As donor** c) Si + Ga : **p-type**, **Ga acceptor**

5. The version of Slide 39 (germanium doped with arsenic)

Arsenic keeps its 5 valence electrons and the host keeps 4:

$$\Delta Z_{\text{As in Ge}} = 5 - 4 = +1 \ \Rightarrow\ \text{donor, n-type} \qquad \frac{E_d}{k_B T} = \frac{0.0127\ \text{eV}}{0.02586\ \text{eV}} = 0.49$$

RESULT & CHECKS

Checks

- **Dimensions.** All energy ratios are eV over eV, dimensionless; all three levels are of the order of $k_B T$ or a few times it, so a large fraction of the impurities is ionised at room temperature.
- **Charge neutrality.** The ionised donor is a fixed positive ion (it lost an electron) and the ionised acceptor a fixed negative ion; the crystal stays neutral.
- **Limit.** For $\Delta Z = 0$ (an isoelectronic atom such as Ge in Si) no carriers are added and the material stays intrinsic.

Interpretation

Sb and As add electrons: in silicon they are donors and the majority carriers are electrons. Ga removes one: it is an acceptor and the majority carriers are holes. In all three cases the dopant is a substitutional atom and contributes one carrier per atom, which is why a small concentration (hundreds of ppm) changes the conductivity by orders of magnitude [Session 13 Slide 26].

Check against the official solution

Item	Official	Computed	Comment
a) Si + Sb	n-type extrinsic	n-type, Sb is a donor	agrees
b) Si + As	n-type extrinsic	n-type, As is a donor	agrees
c) Si + Ga	p-type extrinsic	p-type, Ga is an acceptor	agrees

Problem 6 | Problems T6andT7_Electric and Magnetic Properties.pdf #6 | n-type silicon · conductivity

Phosphorus-Doped Silicon: Type and Conductivity

Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf, p. 2

6. P is added to high purity silicon in order to introduce a concentration of 10^{+23} m^{-3} of charge carriers at room temperature.

- *a) What type of semiconductor is this material?*
- *b) What is the conductivity of this semiconductor at room temperature? Assume that the mobility of the e^- in the material is 0.14 m²/(V·s)*

Official Solution: b) 2243 Ω⁻¹m⁻¹

HYPOTHESES & PARAMETERS

Hypotheses

1. **P is a donor.** Phosphorus (group 15, 5 valence electrons) replaces silicon (4 valence electrons) and gives one electron to the conduction band, so the extrinsic silicon is n-type [Session 13 Slide 36].
2. **Electrons carry the current.** In n-type material $n_e \gg n_H$ and the conductivity is $\sigma \approx n_e|q_e|\mu_e$ [Session 13 Slide 36]. The "concentration of charge carriers" of the statement is the electron concentration n_e.
3. **Holes are negligible.** The holes of the intrinsic silicon are at most $n_i = 1.5 \times 10^{16}$ m⁻³ [Session 13 Slide 31], far fewer than the donor electrons; this is checked in step 5.
4. **Constant of the electron charge.** $|q_e| = 1.602 \times 10^{-19}$ C. Slide 10 prints 1.6×10^{-19} C; the key of the sheet (2243) is reproduced with 1.602×10^{-19} C.

Data

Quantity	Symbol	Value	In SI
Electron (carrier) concentration	n_e	10^{23} m⁻³	10^{23} m⁻³
Electron mobility	μ_e	0.14 m²/(V·s)	0.14 m²/(V·s)
Electron charge	$\lvert q_e \rvert$	1.602×10^{-19} C	same

Unknowns: (a) the type of semiconductor; (b) the conductivity σ.

MODEL & STRATEGY

Every donor atom adds one free electron, so the electron concentration equals the carrier concentration given by the statement. The current density follows from the drift velocity $v_d = \mu_e E$, and the conductivity is the proportionality constant between J and E (Ohm's law of the microscopic form, Slide 12). The figure shows the bar.

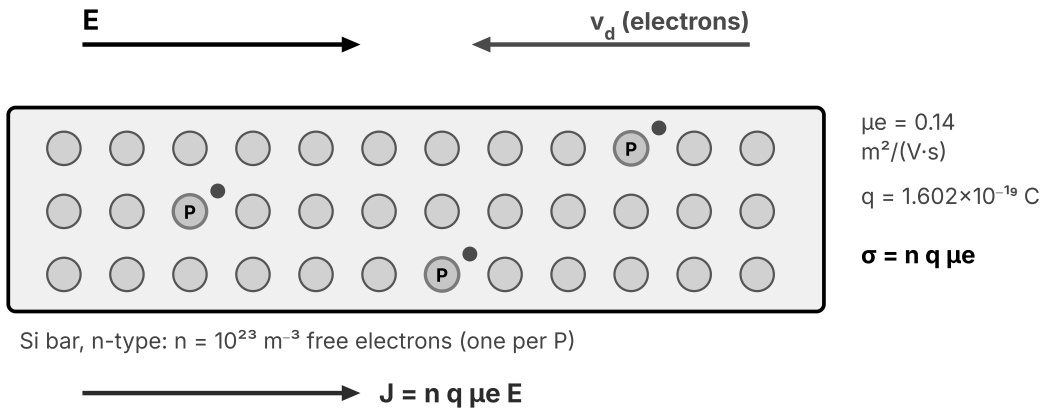


Figure 6. Problem 6: n-type silicon with 10^{23} m^{-3} conduction electrons from P donors; the field E drives the electrons against E, so the current density J points along E (schematic).

Strategy.

1. Classify the semiconductor from the group of P.
2. Write the current density from the drift velocity and read off the conductivity.
3. Evaluate the product in SI units, checking that the units reduce to $\Omega^{-1}\text{m}^{-1}$.
4. Check that the hole contribution is negligible.

STEP-BY-STEP CALCULATION

1. Type of the semiconductor (a) [Session 13 Slide 36]

Valence electrons of the dopant against the host:

$$Z_{\text{P}} - Z_{\text{Si}} = 5 - 4 = +1 \;\Rightarrow\; \text{donor} \;\Rightarrow\; \textbf{n-type extrinsic semiconductor}$$

2. Conductivity from the drift velocity [Session 13 Slides 11-12, 36]

Drift velocity and current density of n_e electrons of charge $|q_e|$, and Ohm's law $J = \sigma E$:

$$v_d = \mu_e \, E \qquad J = n_e \, |q_e| \, v_d = n_e \, |q_e| \, \mu_e \, E \qquad J = \sigma E$$

Comparing the last two expressions of the current density:

$$\sigma = n_e \, |q_e| \, \mu_e$$

3. Numerical evaluation (b)

Charge density of the free electrons, one operation at a time:

$$n_e \, |q_e| = 10^{23}\ \mathrm{m^{-3}} \times 1.602 \times 10^{-19}\ \mathrm{C} = 1.602 \times 10^{4}\ \frac{\mathrm{C}}{\mathrm{m^3}} = 16\,020\ \frac{\mathrm{C}}{\mathrm{m^3}}$$

Multiplication by the mobility:

$$\sigma = 16\,020\ \frac{\mathrm{C}}{\mathrm{m^3}} \times 0.14\ \frac{\mathrm{m^2}}{\mathrm{V\,s}} = 2242.8\ \frac{\mathrm{C}}{\mathrm{V\,s\,m}}$$

Reduction of the units, with $1\ \mathrm{C/s} = 1\ \mathrm{A}$ and $1\ \mathrm{V/A} = 1\ \Omega$:

$$\frac{\mathrm{C}}{\mathrm{V\,s\,m}} = \frac{\mathrm{A}}{\mathrm{V\,m}} = \frac{1}{\Omega\,\mathrm{m}} = \Omega^{-1}\mathrm{m}^{-1}$$

$$\boldsymbol{\sigma = 2243\ \Omega^{-1}\mathrm{m}^{-1}}$$

4. Resistivity (for comparison with the metals of Problem T6-01)

$$\rho = \frac{1}{\sigma} = \frac{1}{2242.8\ \Omega^{-1}\mathrm{m}^{-1}} = 4.459 \times 10^{-4}\ \Omega\,\mathrm{m}$$

5. The holes are negligible

Upper bound of the hole contribution, if there were n_i holes with the hole mobility of silicon (0.048 m²/(V·s) [Session 13 Slide 31]):

$$\sigma_h \leq p\,|q_e|\,\mu_h = 1.5 \times 10^{16}\ \mathrm{m^{-3}} \times 1.602 \times 10^{-19}\ \mathrm{C} \times 0.048\ \frac{\mathrm{m^2}}{\mathrm{V\,s}} = 1.153 \times 10^{-4}\ \Omega^{-1}\mathrm{m}^{-1}$$

$$\frac{\sigma_h}{\sigma} \leq \frac{1.153 \times 10^{-4}}{2242.8} = 5.1 \times 10^{-8}$$

6. Effect of the rounded charge

With the value of Slide 10:

$$\sigma = 10^{23} \times 1.6 \times 10^{-19} \times 0.14\ \Omega^{-1}\mathrm{m}^{-1} = 2240\ \Omega^{-1}\mathrm{m}^{-1}$$

RESULT & CHECKS

Checks

- **Units.** $[n_e|q_e|\mu_e] = \mathrm{m^{-3}} \cdot \mathrm{C} \cdot \mathrm{m^2 V^{-1} s^{-1}} = \Omega^{-1}\mathrm{m}^{-1}$ (step 3).
- **Order of magnitude.** 2243 $\Omega^{-1}\mathrm{m}^{-1}$ lies inside the semiconductor range of Slide 5 (10^{-6} to 10^{4} $\Omega^{-1}\mathrm{m}^{-1}$). Against the intrinsic value of silicon at room temperature, 4×10^{-4} $\Omega^{-1}\mathrm{m}^{-1}$ (the value used in Problem 8 of the sheet and printed on Slide 33; Slide 21 prints 4.5×10^{-4} $\Omega^{-1}\mathrm{m}^{-1}$):

$$\frac{\sigma}{\sigma_i} = \frac{2242.8\ \Omega^{-1}\mathrm{m}^{-1}}{4 \times 10^{-4}\ \Omega^{-1}\mathrm{m}^{-1}} = 5.6 \times 10^{6}$$

- **Hole share.** Below 10^{-7} of the total.
- **Limit.** For $n_e \to n_i$ the formula would need both carriers and tend to the intrinsic value; for $n_e \to 0, \sigma \to 0$.

Interpretation

The donors are a tiny fraction of the silicon atoms. The atom density follows from the density of Slide 31, the molar mass of silicon (28.09 g/mol, periodic table, not on the slides) and Avogadro's number:

$$N_{\mathrm{Si}} = \frac{\rho_m N_A}{M} = \frac{2.33 \times 10^{6}\ \mathrm{g/m^3} \times 6.022 \times 10^{23}\ \mathrm{mol^{-1}}}{28.09\ \mathrm{g/mol}} = 4.995 \times 10^{28}\ \mathrm{m^{-3}}$$

and the fraction of donors is:

$$\frac{n_e}{N_{\mathrm{Si}}} = \frac{10^{23}\ \mathrm{m^{-3}}}{4.995 \times 10^{28}\ \mathrm{m^{-3}}} = 2.0 \times 10^{-6}$$

that is, two donors per million atoms, and yet they raise the conductivity by almost seven orders of magnitude. The mobility of 0.14 m²/(V·s) is the one of the electrons in this material; Slide 42 shows that it falls when the dopant concentration is higher.

Check against the official solution

Item	Official	Computed	Comment
a) Type	not printed (the key lists b only)	n-type extrinsic	P is a group-15 donor, Slide 36
b) Conductivity	2243 Ω⁻¹m⁻¹	2243 Ω⁻¹m⁻¹	agrees with $q = 1.602 \times 10^{-19}$ C; with the rounded 1.6×10^{-19} C of Slide 10 the result is 2240

Problem 7 | Problems T6andT7_Electric and Magnetic Properties.pdf #7 | Intrinsic semiconductor · Arrhenius

Conductivity of Intrinsic Germanium at 75 °C

Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf, p. 2

7. The electrical conductivity of intrinsic germanium at 25 ºC is 2.22 Ω⁻¹m⁻¹. What will its conductivity be at 75 ºC? The band gap of germanium is 0.67 eV and $k_B = 8.62 \cdot 10^{-5}$ *eV/K*

Official Solution: 14.4 Ω⁻¹m⁻¹

HYPOTHESES & PARAMETERS

Hypotheses

1. **Intrinsic law.** In an intrinsic semiconductor the carrier number, and with it the conductivity, follows $\sigma = \sigma_0 \exp(-E_g/2k_BT)$ with the absolute temperature [Session 13 Slide 29]. The constant σ_0 is the same at both temperatures.
2. **Slide approximation.** The slide writes the proportionality $n_i \propto \exp(-E_g/2kT)$ and neglects the slower dependence of the prefactor and of the mobilities on the temperature; the exponential, which varies by orders of magnitude, dominates the ratio.
3. **Constant gap.** $E_g = 0.67$ eV is the same at 25 °C and 75 °C.
4. **Kelvin.** $T[\mathrm{K}] = T[^\circ\mathrm{C}] + 273.15$.

Data

Quantity	Symbol	Value as printed	In SI
Conductivity at 25 °C	σ_1	2.22 Ω⁻¹m⁻¹	2.22 Ω⁻¹m⁻¹
Temperatures	T_1, T_2	25 °C, 75 °C	298.15 K, 348.15 K
Band gap	E_g	0.67 eV	$0.67 \times 1.602 \times 10^{-19}$ J
Boltzmann constant	k_B	8.62×10^{-5} eV/K	$8.62 \times 10^{-5} \times 1.602 \times 10^{-19}$ J/K

Unknown: the conductivity σ_2 at 75 °C.

MODEL & STRATEGY

The unknown constant σ_0 is eliminated by taking the ratio of the law at the two temperatures; what remains is the exponential of $E_g/2k_B$ times the difference of the reciprocal temperatures. The straight line of $\ln\sigma$ against $1/T$ has the slope $-E_g/2k_B$, so the ratio is also the vertical distance between the two points of the figure.

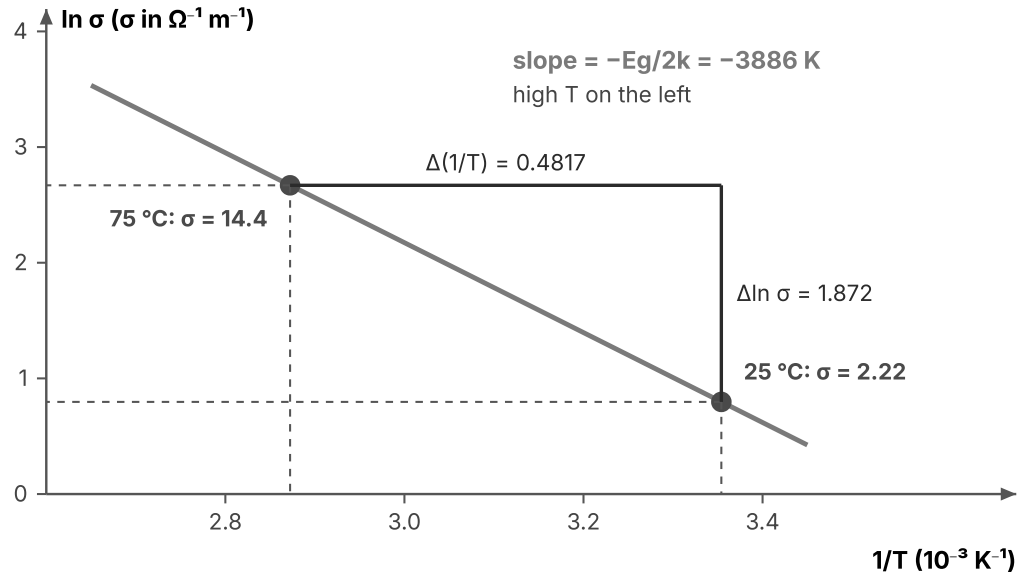


Figure 7. Problem 7: ln σ = ln σ0 − Eg/(2kT) for germanium (Eg = 0.67 eV); the points are the given 25 °C value and the computed 75 °C value, σ = 14.4 Ω⁻¹m⁻¹.

Strategy.

1. Write the law at T_1 and T_2 and divide.
2. Evaluate $E_g/2k_B$ in kelvin, the two reciprocals, their difference and the exponent.
3. Take the exponential and multiply by σ_1.

STEP-BY-STEP CALCULATION

1. The law at both temperatures [Session 13 Slide 29]

$$\sigma_1 = \sigma_0 \exp\left(-\frac{E_g}{2k_B T_1}\right) \qquad \sigma_2 = \sigma_0 \exp\left(-\frac{E_g}{2k_B T_2}\right)$$

2. Elimination of σ_0

Division of the second by the first and the rule $e^a/e^b = e^{a-b}$:

$$\frac{\sigma_2}{\sigma_1} = \frac{\exp\left(-\frac{E_g}{2k_B T_2}\right)}{\exp\left(-\frac{E_g}{2k_B T_1}\right)} = \exp\left[-\frac{E_g}{2k_B T_2} + \frac{E_g}{2k_B T_1}\right]$$

Common factor $E_g/2k_B$:

$$\sigma_2 = \sigma_1 \exp\left[\frac{E_g}{2k_B}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right]$$

3. Temperatures in kelvin

$$T_1 = 25 + 273.15 = 298.15\ \text{K} \qquad T_2 = 75 + 273.15 = 348.15\ \text{K}$$

4. The constant $E_g/2k_B$

Direct, with both quantities in eV (the eV cancels):

$$\frac{E_g}{2k_B} = \frac{0.67\ \text{eV}}{2 \times 8.62 \times 10^{-5}\ \text{eV/K}} = \frac{0.67}{1.724 \times 10^{-4}}\ \text{K} = 3886.3\ \text{K}$$

The same result in joules, as a check of the unit conversion ($1\ \text{eV} = 1.602 \times 10^{-19}$ J):

$$E_g = 0.67\ \text{eV} \times 1.602 \times 10^{-19}\ \frac{\text{J}}{\text{eV}} = 1.0733 \times 10^{-19}\ \text{J} \qquad k_B = 8.62 \times 10^{-5}\ \frac{\text{eV}}{\text{K}} \times 1.602 \times 10^{-19}\ \frac{\text{J}}{\text{eV}} = 1.3809 \times 10^{-23}\ \frac{\text{J}}{\text{K}}$$

$$\frac{E_g}{2k_B} = \frac{1.0733 \times 10^{-19}\ \text{J}}{2 \times 1.3809 \times 10^{-23}\ \text{J/K}} = 3886.3\ \text{K}$$

5. Reciprocal temperatures and their difference

$$\frac{1}{T_1} = \frac{1}{298.15\ \text{K}} = 3.35402 \times 10^{-3}\ \text{K}^{-1} \qquad \frac{1}{T_2} = \frac{1}{348.15\ \text{K}} = 2.87233 \times 10^{-3}\ \text{K}^{-1}$$

$$\frac{1}{T_1} - \frac{1}{T_2} = 3.35402 \times 10^{-3}\ \text{K}^{-1} - 2.87233 \times 10^{-3}\ \text{K}^{-1} = 4.8169 \times 10^{-4}\ \text{K}^{-1}$$

6. Exponent, exponential and result

$$\frac{E_g}{2k_B}\left(\frac{1}{T_1}-\frac{1}{T_2}\right) = 3886.3\ \mathrm{K}\times 4.8169\times 10^{-4}\ \mathrm{K^{-1}} = 1.8720$$

$$\frac{\sigma_2}{\sigma_1} = e^{1.8720} = 6.501$$

$$\sigma_2 = 2.22\ \Omega^{-1}\mathrm{m}^{-1}\times 6.501 = \mathbf{14.43\ \Omega^{-1}m^{-1}}$$

7. Using 273 instead of 273.15

$$\frac{1}{298}-\frac{1}{348} = 4.8214\times 10^{-4}\ \mathrm{K^{-1}} \qquad \sigma_2 = 2.22\times\exp(3886.3\times 4.8214\times 10^{-4}) = 2.22\times e^{1.8737} = 14.46\ \Omega^{-1}\mathrm{m}^{-1}$$

RESULT & CHECKS

Checks

- **Dimensionless exponent.**

$$\left[\frac{E_g}{2k_B}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)\right] = \mathrm{K}\cdot\mathrm{K^{-1}} = 1$$

- **Slope.** On the plot of $\ln\sigma$ against $1/T$ the line has the slope -3886.3 K, and the vertical distance between the two points is $\ln(\sigma_2/\sigma_1) = 1.8720$.

$$\ln\frac{\sigma_2}{\sigma_1} = \ln\frac{14.43}{2.22} = \ln 6.501 = 1.8720$$

- **Consistency with the slide data.** Slide 31 gives $n_i = 2.4 \times 10^{19}$ m⁻³, $\mu_e = 0.39$ and $\mu_h = 0.19$ m²/(V·s) for germanium at room temperature:

$$\sigma_1 = n_i \, |q_e| \, (\mu_e + \mu_h) = 2.4 \times 10^{19} \times 1.602 \times 10^{-19} \times 0.58 \; \Omega^{-1}\mathrm{m}^{-1} = 2.23 \; \Omega^{-1}\mathrm{m}^{-1}$$

which agrees with the 2.22 of the statement.

Interpretation

A rise of 50 K multiplies the conductivity of germanium by 6.5: in a semiconductor heat creates carriers. At the limit $T_2 \to T_1$ the ratio tends to 1; at the limit $T_2 \to \infty$ the ratio saturates at the value of the next block (within the model of constant prefactor):

$$\frac{E_g}{2k_B T_1} = \frac{3886.3 \text{ K}}{298.15 \text{ K}} = 13.035 \qquad e^{13.035} = 4.58 \times 10^5$$

Check against the official solution

Item	Official	Computed	Comment
Conductivity at 75 °C	14.4 Ω⁻¹m⁻¹	14.43 Ω⁻¹m⁻¹	agrees with 273.15 K; with 273 K the result is 14.46, which rounds to 14.5, so the key uses 273.15

Problem 8	Problems T6andT7_Electric and Magnetic Properties.pdf #8	Intrinsic carriers · electrons and holes

Carrier Concentrations of Intrinsic Silicon from its Conductivity

Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf, p. 2

8. The electrical conductivity of intrinsic Si at room temperature is $4 \cdot 10^{-4}\ \Omega^{-1}m^{-1}$, and the mobilities of electrons and holes are 0.14 y 0.048 m²/(V·s) respectively. Calculate the concentration of holes and electrons.

The statement is transcribed as printed: "y" is the Spanish "and".

Official Solution: 1.32×10^{16} m⁻³

HYPOTHESES & PARAMETERS

Hypotheses

1. **Two carrier types.** In an intrinsic semiconductor each excitation creates one electron in the conduction band and one hole in the valence band, so both carry current: $\sigma = n|q_e|\mu_e + p|q_e|\mu_h$ [Session 13 Slides 27-28].
2. **Equal numbers.** Because the carriers are created in pairs, $n = p = n_i$ and the charge of the hole has the same magnitude as that of the electron, $|q_h| = |q_e|$ [Session 13 Slide 28].
3. **Room temperature,** all values are those of the statement. The charge is 1.602×10^{-19} C (the value that reproduces the key).

Data

Quantity	Symbol	Value	In SI
Conductivity	σ	4×10^{-4} Ω⁻¹m⁻¹	same
Electron mobility	μ_e	0.14 m²/(V·s)	same
Hole mobility	μ_h	0.048 m²/(V·s)	same
Charge	$\lvert q_e \rvert$	1.602×10^{-19} C	same

Unknown: the hole concentration p and the electron concentration n.

MODEL & STRATEGY

The two carriers add their currents. As their numbers are equal, the conductivity is the carrier concentration times the charge times the sum of the two mobilities, and that one equation has one unknown. The figure shows the excitation of pairs and the way the conductivity splits between the carriers.

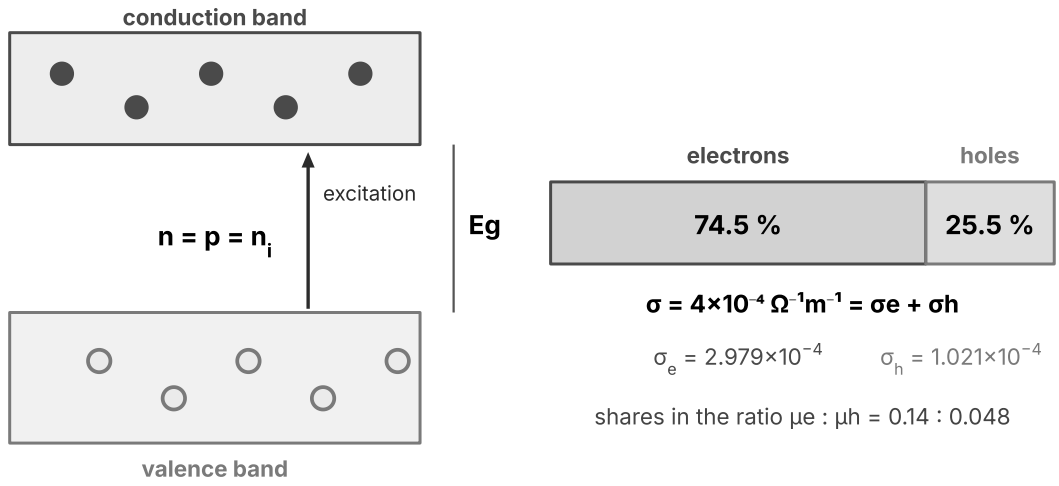


Figure 8. Problem 8: an intrinsic semiconductor has as many electrons as holes (n = p = ni, the intrinsic concentration); the conductivity splits in the ratio of the mobilities, 0.14 to 0.048 (schematic band diagram, bar to scale).

Strategy.

1. Write the conductivity with both carriers and impose $n = p$.
2. Solve for the concentration.
3. Evaluate with SI values, then compute the share of each carrier.
4. Compare with the intrinsic concentration of silicon tabulated on Slide 31.

STEP-BY-STEP CALCULATION

1. Conductivity with both carriers [Session 13 Slides 27-28]

$$\sigma = n\,|q_e|\,\mu_e + p\,|q_e|\,\mu_h$$

2. Condition of pairs and solution for the concentration

Substitution of $n = p = n_i$ and extraction of the common factor:

$$\sigma = n_i\,|q_e|\,\mu_e + n_i\,|q_e|\,\mu_h = n_i\,|q_e|\,(\mu_e + \mu_h)$$

Division by $|q_e|(\mu_e + \mu_h)$:

$$n_i = \frac{\sigma}{|q_e|\,(\mu_e + \mu_h)}$$

3. Numerical evaluation, one operation per block

Sum of the mobilities:

$$\mu_e + \mu_h = 0.14\ \frac{\mathrm{m}^2}{\mathrm{V\,s}} + 0.048\ \frac{\mathrm{m}^2}{\mathrm{V\,s}} = 0.188\ \frac{\mathrm{m}^2}{\mathrm{V\,s}}$$

Denominator:

$$|q_e|\,(\mu_e + \mu_h) = 1.602 \times 10^{-19}\ \mathrm{C} \times 0.188\ \frac{\mathrm{m}^2}{\mathrm{V\,s}} = 3.01176 \times 10^{-20}\ \frac{\mathrm{C\,m}^2}{\mathrm{V\,s}}$$

Quotient:

$$n_i = \frac{4 \times 10^{-4}\ \Omega^{-1}\mathrm{m}^{-1}}{3.01176 \times 10^{-20}\ \mathrm{C\,m^2 V^{-1} s^{-1}}} = 1.3281 \times 10^{16}\ \frac{\Omega^{-1}\mathrm{m}^{-1}\,\mathrm{V\,s}}{\mathrm{C\,m}^2}$$

Reduction of the units, with $\Omega^{-1} = \mathrm{A/V}$ and $\mathrm{A\,s} = \mathrm{C}$:

$$\frac{\Omega^{-1}\mathrm{m}^{-1}\,\mathrm{V\,s}}{\mathrm{C\,m}^2} = \frac{\mathrm{A\,V\,s}}{\mathrm{V\,C\,m}^3} = \frac{\mathrm{C}}{\mathrm{C\,m}^3} = \mathrm{m}^{-3}$$

$$\mathbf{n = p = n_i = 1.328 \times 10^{16}\ m^{-3}}$$

4. Share of each carrier in the conductivity

Electron and hole conductivities:

$$\sigma_e = n\,|q_e|\,\mu_e = 1.3281\times10^{16}\times1.602\times10^{-19}\times0.14\,\Omega^{-1}\mathrm{m}^{-1} = 2.979\times10^{-4}\,\Omega^{-1}\mathrm{m}^{-1}$$

$$\sigma_h = p\,|q_e|\,\mu_h = 1.3281\times10^{16}\times1.602\times10^{-19}\times0.048\,\Omega^{-1}\mathrm{m}^{-1} = 1.021\times10^{-4}\,\Omega^{-1}\mathrm{m}^{-1}$$

Shares (they equal the mobility ratios because $n = p$):

$$\frac{\sigma_e}{\sigma} = \frac{0.14}{0.188} = 0.7447 \qquad \frac{\sigma_h}{\sigma} = \frac{0.048}{0.188} = 0.2553$$

Sum:

$$\sigma_e + \sigma_h = 2.979\times10^{-4} + 1.021\times10^{-4} = 4.000\times10^{-4}\,\Omega^{-1}\mathrm{m}^{-1}$$

5. Comparison with the table of Slide 31

Slide 31 gives for silicon $n_i = 1.5\times10^{16}$ m⁻³, $\mu_e = 0.135$ and $\mu_h = 0.048$ m²/(V·s):

$$\sigma = 1.5\times10^{16}\times1.602\times10^{-19}\times(0.135+0.048)\,\Omega^{-1}\mathrm{m}^{-1} = 4.40\times10^{-4}\,\Omega^{-1}\mathrm{m}^{-1}$$

$$\frac{1.328\times10^{16}}{1.5\times10^{16}} = 0.885$$

RESULT & CHECKS

Checks

- **Units.** $\sigma/(|q|\mu)$ has units of m^{-3} (step 3).
- **Reconstruction.** The two partial conductivities add up to the given 4×10^{-4} $\Omega^{-1}\mathrm{m}^{-1}$ (step 4).
- **Order of magnitude.** Relative difference of the computed concentration from the tabulated 1.5×10^{16} m^{-3}; it comes from the rounded conductivity of the statement (4×10^{-4}, with $\mu_e = 0.14$) against the 4.4×10^{-4} that the table values give:

$$\frac{1.328 \times 10^{16} - 1.5 \times 10^{16}}{1.5 \times 10^{16}} = -0.115$$

- **Concentration against atom density.** With $N_{\mathrm{Si}} = 4.995 \times 10^{28}$ atoms/m³ (from $\rho_m N_A / M$, Problem T6-09):

$$\frac{N_{\mathrm{Si}}}{n_i} = \frac{4.995 \times 10^{28}\ \mathrm{m}^{-3}}{1.328 \times 10^{16}\ \mathrm{m}^{-3}} = 3.76 \times 10^{12}$$

one carrier pair per 3.8×10^{12} atoms, which is why intrinsic silicon is such a poor conductor.

Interpretation

Silicon at room temperature has 1.3×10^{16} electrons and the same number of holes per cubic metre, and three quarters of the current is carried by the electrons because they move almost three times faster. If the holes had the mobility of the electrons the two shares would be equal; if the number of electrons exceeded that of the holes (doping) the material would no longer be intrinsic.

Check against the official solution

Item	Official	Computed	Comment
Concentration of holes and electrons	1.32×10^{16} m^{-3}	1.328×10^{16} m^{-3} for each	agrees to 0.6 %, see the warning

DISCREPANCY WITH THE OFFICIAL SOLUTION

The key prints 1.32×10^{16} m^{-3}. Our exact value is 1.3281×10^{16} m^{-3}, which rounds to 1.33×10^{16}: the key appears to truncate instead of rounding. With the rounded charge 1.6×10^{-19} C the result would be 1.330×10^{16} m^{-3}. We keep the computed value.

Problem 9 | Problems T6andT7_Electric and Magnetic Properties.pdf #9 | Ranking · temperature · doping · concentration

Ranking Semiconductors by Conductivity: Temperature, Doping and Concentration

Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf, p. 2

9. Order the following semiconductors according to their conductivity (assume that the materials are always at room temperature, except when the opposite is specified).

- *a) Pure Germanium at 0 ºC; pure Germanium at 25 ºC; pure Germanium at 50 ºC.*
- *b) Phosphor doped- silicon; pure silicon.*
- *c) Germanium doped with 100 ppb of nitrogen; germanium doped with 1ppm of nitrogen.*

The sheet prints no solution for this problem.

HYPOTHESES & PARAMETERS

Hypotheses

1. **Pure germanium is intrinsic** and obeys $\sigma = \sigma_0 \exp(-E_g/2k_BT)$ [Session 13 Slide 29]; σ_0 does not change over 50 K, and $E_g = 0.67$ eV [Session 13 Slide 31].
2. **Doped silicon is extrinsic.** Phosphorus is a group-15 donor, so $\sigma \approx n_e|q_e|\mu_e$ with n_e equal to the number of P atoms [Session 13 Slide 36]. Extrinsic doping is "generally 100-1000 ppm" [Session 13 Slide 26]; a doping of 1 ppm, which is 100 times below the lowest value of the slide, is used so that the conclusion holds a fortiori for any realistic doping. The doping level is not given in the statement. A concentration in ppm is read as an atomic fraction (atoms of dopant per million atoms of host).
3. **Mobility of the doped silicon.** It lies between the value of low doping, 0.135 m²/(V·s) [Session 13 Slide 31], and a floor of 0.001 m²/(V·s), the bottom of the axis of the mobility chart of Slide 42, which the electron curve never reaches.
4. **Nitrogen is a donor** in the course model, because it is in group 15 (VA), like P, As and Sb [Session 13 Slides 35-36]. Extension caveat: in real germanium nitrogen is not a practical donor; the problem is answered within the rule of the course.
5. **Same mobility** for the two nitrogen levels, so $\sigma \propto n_e$: for extrinsic semiconductors the distortion of the lattice caused by the doping is negligible, and the conductivity is controlled by the number of charge carriers [Session 13 Slide 42].
6. **Room temperature** is 25 °C (298.15 K); $k_B = 8.62 \times 10^{-5}$ eV/K; $|q_e| = 1.602 \times 10^{-19}$ C. The atomic masses (Si 28.09 g/mol, Ge 72.63 g/mol) and Avogadro's number (6.022×10^{23} mol⁻¹) are standard data, not on the slides.

Data

Quantity	Value	Source
E_g of Ge	0.67 eV	Slide 31
Density of Si, Ge	2.33×10^6, 5.32×10^6 g/m³	Slide 31
Intrinsic conductivity of Si	4×10^{-4} Ω⁻¹m⁻¹	Slides 31, 33
Intrinsic carrier density of Ge	2.4×10^{19} m⁻³	Slide 31
μ_e of Si at low doping	0.135 m²/(V·s)	Slide 31

Unknown: the order of the conductivities in each group.

MODEL & STRATEGY

Each group changes one variable: the temperature (a), the type of material (b) or the doping concentration (c). The figure collects the numerical results of the three groups.

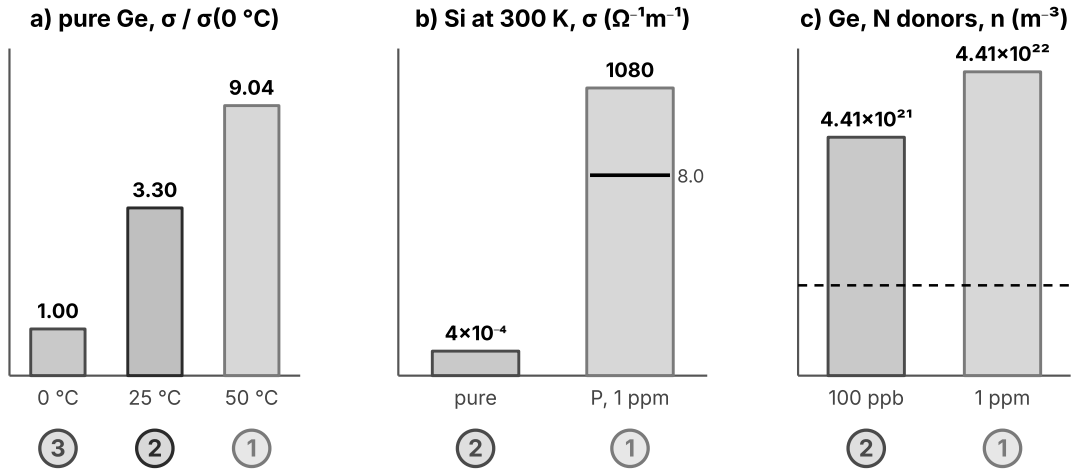


Figure 9. Problem 9: (a) germanium against temperature, (b) pure and phosphorus-doped silicon, (c) germanium with 100 ppb and 1 ppm of N; bars are computed with the data of the solution (logarithmic scale in b and c), 1 marks the highest conductivity.

Strategy.

1. (a) Compute the ratios of the intrinsic conductivity at the three temperatures from the exponential law.
2. (b) Compute the conductivity of the doped silicon at its lowest realistic doping and compare it with the intrinsic value.
3. (c) Convert both concentrations to carriers per cubic metre and compare them with each other and with the intrinsic density.

STEP-BY-STEP CALCULATION

a) Pure germanium at 0, 25 and 50 °C [Session 13 Slide 29]

Ratio between two temperatures (the derivation of Problem T6-07), and the temperatures in kelvin:

$$\frac{\sigma_B}{\sigma_A} = \exp\left[\frac{E_g}{2k_B}\left(\frac{1}{T_A} - \frac{1}{T_B}\right)\right] \qquad \frac{E_g}{2k_B} = \frac{0.67\ \text{eV}}{2 \times 8.62 \times 10^{-5}\ \text{eV/K}} = 3886.3\ \text{K}$$

$$T_0 = 0 + 273.15 = 273.15\ \text{K} \qquad T_{25} = 25 + 273.15 = 298.15\ \text{K} \qquad T_{50} = 50 + 273.15 = 323.15\ \text{K}$$

Reciprocals:

$$\frac{1}{T_0} = 3.66099 \times 10^{-3}\ \text{K}^{-1} \qquad \frac{1}{T_{25}} = 3.35402 \times 10^{-3}\ \text{K}^{-1} \qquad \frac{1}{T_{50}} = 3.09454 \times 10^{-3}\ \text{K}^{-1}$$

Exponents and ratios for 0 to 25 °C, 25 to 50 °C and 0 to 50 °C:

$$\frac{\sigma_{25}}{\sigma_0} = \exp\left[3886.3\ \text{K} \times (3.66099 - 3.35402) \times 10^{-3}\ \text{K}^{-1}\right] = \exp(1.1930) = 3.297$$

$$\frac{\sigma_{50}}{\sigma_{25}} = \exp\left[3886.3\ \text{K} \times (3.35402 - 3.09454) \times 10^{-3}\ \text{K}^{-1}\right] = \exp(1.0084) = 2.741$$

$$\frac{\sigma_{50}}{\sigma_0} = 3.297 \times 2.741 = 9.038 = \exp(2.2014)$$

$$\sigma(\mathbf{50}\ ^\circ\mathbf{C}) > \sigma(\mathbf{25}\ ^\circ\mathbf{C}) > \sigma(\mathbf{0}\ ^\circ\mathbf{C})$$

b) Phosphorus-doped silicon against pure silicon [Session 13 Slides 26, 31, 33, 36]

Atomic density of silicon, with $N = \rho_m N_A / M$:

$$N_{\mathrm{Si}} = \frac{2.33 \times 10^{6}\ \mathrm{g/m^{3}} \times 6.022 \times 10^{23}\ \mathrm{mol^{-1}}}{28.09\ \mathrm{g/mol}} = \frac{1.4031 \times 10^{30}}{28.09}\ \mathrm{m^{-3}} = 4.995 \times 10^{28}\ \mathrm{m^{-3}}$$

Conversion of 1 ppm (one atom in 10^6) to a concentration of donors, one electron per P atom:

$$n_e = 10^{-6} \times 4.995 \times 10^{28}\ \mathrm{m^{-3}} = 4.995 \times 10^{22}\ \mathrm{m^{-3}}$$

Charge density of the free electrons:

$$n_e\,|q_e| = 4.995 \times 10^{22}\ \mathrm{m^{-3}} \times 1.602 \times 10^{-19}\ \mathrm{C} = 8.002 \times 10^{3}\ \frac{\mathrm{C}}{\mathrm{m^{3}}}$$

Conductivity, upper and lower bounds by the mobility ($\sigma = n_e |q_e| \mu_e$, units $\mathrm{C\,m^{-3} \cdot m^2 V^{-1} s^{-1}} = \Omega^{-1}\mathrm{m}^{-1}$):

$$\sigma_{\max} = 8.002 \times 10^{3} \times 0.135\ \Omega^{-1}\mathrm{m}^{-1} = 1080\ \Omega^{-1}\mathrm{m}^{-1} \qquad \sigma_{\min} = 8.002 \times 10^{3} \times 0.001\ \Omega^{-1}\mathrm{m}^{-1} = 8.0\ \Omega^{-1}\mathrm{m}^{-1}$$

Comparison with the intrinsic silicon of Slide 33:

$$\frac{\sigma_{\min}}{\sigma_{\mathrm{pure}}} = \frac{8.0}{4 \times 10^{-4}} = 2.0 \times 10^{4} \qquad \frac{\sigma_{\max}}{\sigma_{\mathrm{pure}}} = \frac{1080}{4 \times 10^{-4}} = 2.7 \times 10^{6}$$

$$\sigma(\text{P-doped Si}) > \sigma(\text{pure Si})$$

A doping of 100 to 1000 ppm, as on Slide 26, is 100 to 1000 times the level used above, so n_e is larger by the same factor and the gap widens further.

c) Germanium with 100 ppb and with 1 ppm of nitrogen

Atomic density of germanium:

$$N_{\mathrm{Ge}} = \frac{5.32 \times 10^{6}\ \mathrm{g/m^{3}} \times 6.022 \times 10^{23}\ \mathrm{mol^{-1}}}{72.63\ \mathrm{g/mol}} = \frac{3.2037 \times 10^{30}}{72.63}\ \mathrm{m^{-3}} = 4.411 \times 10^{28}\ \mathrm{m^{-3}}$$

Conversion of the two concentrations (1 ppm is 10^{-6}, 100 ppb is 10^{-7}), one electron per N atom (hypothesis 4):

$$n_{1\ \mathrm{ppm}} = 10^{-6} \times 4.411 \times 10^{28}\ \mathrm{m^{-3}} = 4.411 \times 10^{22}\ \mathrm{m^{-3}} \qquad n_{100\ \mathrm{ppb}} = 10^{-7} \times 4.411 \times 10^{28}\ \mathrm{m^{-3}} = 4.411 \times 10^{21}\ \mathrm{m^{-3}}$$

Ratio of conductivities, with the same mobility:

$$\frac{\sigma_{1\ \mathrm{ppm}}}{\sigma_{100\ \mathrm{ppb}}} = \frac{n_{1\ \mathrm{ppm}}\,|q_e|\,\mu_e}{n_{100\ \mathrm{ppb}}\,|q_e|\,\mu_e} = \frac{4.411 \times 10^{22}}{4.411 \times 10^{21}} = 10$$

Both are extrinsic, because both exceed the intrinsic density of Slide 31:

$$\frac{n_{100\ \mathrm{ppb}}}{n_i} = \frac{4.411 \times 10^{21}}{2.4 \times 10^{19}} = 183.8 \qquad \frac{n_{1\ \mathrm{ppm}}}{n_i} = \frac{4.411 \times 10^{22}}{2.4 \times 10^{19}} = 1838$$

$$\sigma(\mathbf{1\ ppm\ N}) > \sigma(\mathbf{100\ ppb\ N})$$

Result table

Group	Ranking, highest conductivity first
a	Ge at 50 °C > Ge at 25 °C > Ge at 0 °C
b	P-doped Si > pure Si
c	Ge with 1 ppm N > Ge with 100 ppb N

RESULT & CHECKS

Checks

- **Dimensionless exponents.** $[(E_g/2k_B)(1/T_A - 1/T_B)] = \mathrm{K}\cdot\mathrm{K}^{-1} = 1$.
- **Units of the atomic density.** $[\rho_m N_A/M] = \mathrm{g\,m^{-3}}\cdot\mathrm{mol^{-1}}/(\mathrm{g\,mol^{-1}}) = \mathrm{m^{-3}}$.
- **Consistency.** The result of a for 25 to 50 °C (2.74) is the same exponential law as in Problem T6-07, where the 25 to 75 °C ratio is 6.50.
- **Doping against the intrinsic density of silicon** (Slide 31):

$$\frac{n_e}{n_i} = \frac{4.995\times10^{22}\ \mathrm{m^{-3}}}{1.5\times10^{16}\ \mathrm{m^{-3}}} = 3.3\times10^{6}$$

so one part per million is already far above the intrinsic density.

Interpretation

Heating creates carriers in the pure semiconductor, doping supplies them at room temperature, and the more dopant the more carriers, until the mobility starts to drop (Slide 42). Doping is by far the strongest lever: a one-in-a-million donor does more for the conductivity than a 50 K rise of the pure crystal. The ranking in c assumes that nitrogen behaves as a donor, as the course rule says for group 15; the ratio 10 is only the ratio of the concentrations and does not depend on that assumption except for the sign.

Check against the official solution

Item	Official	Computed	Comment
a	not printed on the sheet	50 °C > 25 °C > 0 °C	exponential law of Slide 29
b	not printed on the sheet	doped Si > pure Si	at least 2×10^4 times
c	not printed on the sheet	1 ppm > 100 ppb	ratio 10

Problem 10	Problems T6andT7_Electric and Magnetic Properties.pdf #10	Ranking · band gap · donor and acceptor levels

Ranking Semiconductors by Conductivity: Band Gap and Donor and Acceptor Levels

Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf, p. 2

10. Order the following semiconductors according to their conductivity (assume that the materials are always at the same temperature).

- *a) Gallium arsenide; aluminium phosphide; Indium antimonide.*
- *b) Cadmium telluride; Indium antimonide.*
- *c) Germanium with 1ppm of P; Germanium with 1 ppm of Sb.*
- *d) Germanium with 1ppm of P; Silicon with 1 ppm of P.*
- *e) Silicon with 10 ppm of B; Silicon with 10 ppm of In.*
- *f) Germanium with 5ppm of As; Germanium with 5 ppm of Sb.*

The sheet prints no solution for this problem. Slides 34 and 41 pose the same items (a, e and f) and ask to "explain the criterion you have followed".

HYPOTHESES & PARAMETERS

Hypotheses

1. **Intrinsic compounds (a, b).** Their conductivity is set by the band gap through $\sigma \propto \exp(-E_g/2k_BT)$ [Session 13 Slide 29]; a smaller E_g gives more excited carriers. The trend of E_g follows the bonding: as the covalent character decreases (the bond becomes more ionic) E_g increases and σ decreases [Session 13 Slide 32]; and within a group E_g decreases as the atom size increases [Session 13 Slide 31].
2. **Extrinsic semiconductors (c to f).** The doping concentration is the same in each pair, so the carriers come from the ionisation of the dopants; the smaller the ionisation energy ($E_c - E_d$ for donors, $E_a - E_v$ for acceptors), the larger the fraction ionised and the higher the conductivity: $\sigma = \sigma_0 \exp[-(E_c - E_d)/2k_BT]$ [Session 13 Slide 43]. The levels are those of Slide 40; for donors and acceptors "increasing σ" runs with increasing ionic radius for donors and decreasing ionic radius for acceptors (arrows of Slide 40).
3. **Host and mobility.** The mobility of the host matters when the two hosts differ (d): μ_e is 0.39 m²/(V·s) in germanium and 0.135 in silicon [Session 13 Slide 31], and in extrinsic material σ is controlled by the number of carriers because the distortion by the doping is negligible [Session 13 Slide 42].
4. **Data outside the slides** are marked as an extension: AlP and CdTe are not tabulated on Slides 31 and 33.
5. $k_B = 8.62 \times 10^{-5}$ eV/K and $T = 300$ K, so $k_BT = 0.02586$ eV.

Data

Item	Quantity	Value (eV)	Source
InSb	E_g	0.17	Slide 33
GaAs	E_g	1.42	Slide 33
GaP (for the trend)	E_g	2.25	Slide 33
AlP	E_g	about 2.45	literature, extension
CdTe	E_g	about 1.5	literature, extension
Ge:P, Ge:As, Ge:Sb	E_d	0.0120, 0.0127, 0.0096	Slide 40
Si:P, Si:As, Si:Sb	E_d	0.045, 0.049, 0.039	Slide 40
Si:B, Si:Al, Si:Ga, Si:In	E_a	0.045, 0.057, 0.065, 0.160	Slide 40

Unknown: the order of the conductivities in each pair or triple.

MODEL & STRATEGY

Items a and b compare band gaps; items c to f compare ionisation energies. The figure puts the two kinds of quantity side by side, with the ranking found.

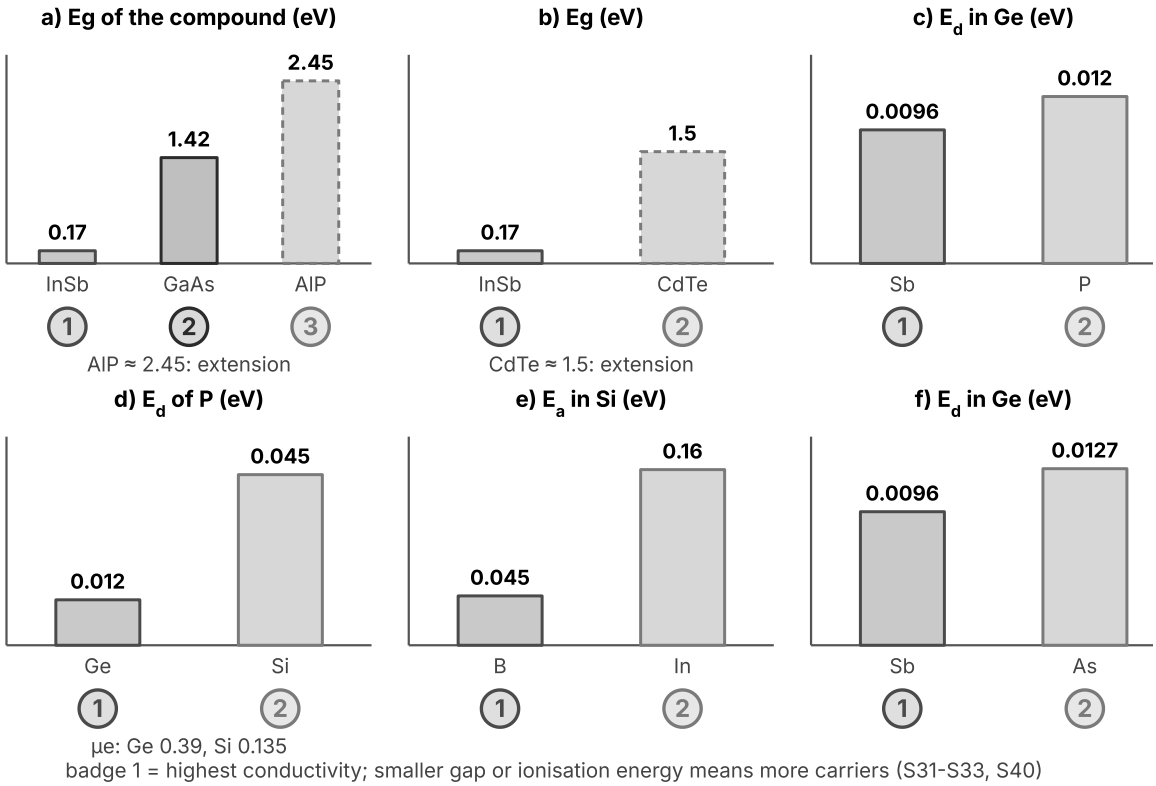


Figure 10. Problem 10: band gaps (a, b) and donor or acceptor ionisation energies (c to f) of the pairs compared, with the data of Session 13 Slides 33 and 40; 1 marks the higher conductivity.

Strategy.

1. (a, b) Read the gaps of the tabulated compounds and place the others with the bonding trend of Slides 31-32.
2. (c to f) Read the donor or acceptor level of each dopant in its host from Slide 40 and apply the order of the exponential.
3. Quantify each comparison with the exponential of Slide 29 or 43 at 300 K, as an indication only: σ_0 is taken equal, and the real extrinsic conductivity saturates when all the dopants are ionised.

STEP-BY-STEP CALCULATION

a) GaAs, AlP and InSb [Session 13 Slides 31-33]

Law and the thermal factor at 300 K:

$$\sigma \propto \exp\left(-\frac{E_g}{2k_BT}\right) \qquad 2k_BT = 2 \times 8.62 \times 10^{-5}\ \frac{\text{eV}}{\text{K}} \times 300\ \text{K} = 0.05172\ \text{eV}$$

Exponents $E_g/2k_BT$ of the three materials:

$$\frac{0.17\ \text{eV}}{0.05172\ \text{eV}} = 3.287\ (\text{InSb}) \qquad \frac{1.42\ \text{eV}}{0.05172\ \text{eV}} = 27.46\ (\text{GaAs}) \qquad \frac{2.45\ \text{eV}}{0.05172\ \text{eV}} = 47.37\ (\text{AlP, extension})$$

Relative factors:

$$e^{-3.287} = 3.74 \times 10^{-2} \qquad e^{-27.46} = 1.19 \times 10^{-12} \qquad e^{-47.37} = 2.67 \times 10^{-21}$$

The three compounds are III-V (covalent with some ionic character); AlP has the smallest atoms of the three (Al and P are in the third period), InSb the largest (fifth period). Larger atoms mean a smaller gap (Slide 31) and the gap is smallest in InSb; on the tabulated side GaAs (1.42 eV) is below GaP (2.25 eV), and AlP, with smaller atoms than GaP, lies above it:

$$E_g(\text{InSb}) < E_g(\text{GaAs}) < E_g(\text{GaP}) < E_g(\text{AlP})$$

$$\sigma(\mathbf{InSb}) > \sigma(\mathbf{GaAs}) > \sigma(\mathbf{AlP})$$

b) CdTe and InSb [Session 13 Slides 32-33]

CdTe is a II-VI compound, more ionic than the III-V InSb; a lower covalent character means a larger gap and a lower conductivity [Session 13 Slide 32]. With the literature value 1.5 eV (extension) and the 0.17 eV of Slide 33, ratio of the factors:

$$\frac{\sigma_{\mathrm{InSb}}}{\sigma_{\mathrm{CdTe}}} \sim \exp\left[\frac{(1.5 - 0.17)\ \mathrm{eV}}{0.05172\ \mathrm{eV}}\right] = \exp(25.72) = 1.5 \times 10^{11}$$

$$\sigma(\mathbf{InSb}) > \sigma(\mathbf{CdTe})$$

c) Germanium with P or Sb [Session 13 Slides 40-41, 43]

Same concentration, so the dopant with the lower donor level ionises more:

$$E_d(\mathrm{Sb}) = 0.0096\ \mathrm{eV} < E_d(\mathrm{P}) = 0.0120\ \mathrm{eV}$$

Indicative ratio, from the exponential of Slide 43 with $2k_BT = 0.05172$ eV and equal σ_0:

$$\frac{\sigma_{\mathrm{Sb}}}{\sigma_{\mathrm{P}}} \sim \exp\left[\frac{(0.0120 - 0.0096)\ \mathrm{eV}}{0.05172\ \mathrm{eV}}\right] = \exp(0.0464) = 1.047$$

$$\sigma(\mathbf{Ge{:}Sb}) > \sigma(\mathbf{Ge{:}P})$$

d) Germanium or silicon with 1 ppm of P

The same dopant in two hosts: Ge has the lower donor level and the higher electron mobility:

$$E_d(\mathrm{Ge}) = 0.0120\ \mathrm{eV} < E_d(\mathrm{Si}) = 0.045\ \mathrm{eV} \qquad \mu_e(\mathrm{Ge}) = 0.39\ \frac{\mathrm{m^2}}{\mathrm{V\,s}} > \mu_e(\mathrm{Si}) = 0.135\ \frac{\mathrm{m^2}}{\mathrm{V\,s}}$$

Indicative ratio, the ionisation exponential times the mobility ratio:

$$\frac{\sigma_{\mathrm{Ge}}}{\sigma_{\mathrm{Si}}} \sim \exp\!\left[\frac{(0.045 - 0.0120)\ \mathrm{eV}}{0.05172\ \mathrm{eV}}\right] \times \frac{0.39}{0.135} = \exp(0.6380) \times 2.889 = 1.893 \times 2.889 = 5.47$$

The same atomic fraction of 1 ppm is a slightly smaller number of donors per unit volume in germanium than in silicon, because germanium has fewer atoms per cubic metre ($N_{\mathrm{Ge}} = 4.411 \times 10^{28}$ m⁻³ and $N_{\mathrm{Si}} = 4.995 \times 10^{28}$ m⁻³, from the densities of Slide 31 as in Problem 9). Including that factor:

$$\frac{N_{\mathrm{Ge}}}{N_{\mathrm{Si}}} = \frac{4.411 \times 10^{28}\ \mathrm{m^{-3}}}{4.995 \times 10^{28}\ \mathrm{m^{-3}}} = 0.8831 \qquad \frac{\sigma_{\mathrm{Ge}}}{\sigma_{\mathrm{Si}}} \sim 5.47 \times 0.8831 = 4.83$$

The order does not change.

$$\sigma(\mathbf{Ge{:}P}) > \sigma(\mathbf{Si{:}P})$$

e) Silicon with 10 ppm of B or In

Same concentration of acceptors; the acceptor level of B is much closer to the valence band:

$$E_a(\mathrm{B}) = 0.045\ \mathrm{eV} < E_a(\mathrm{In}) = 0.160\ \mathrm{eV}$$

Indicative ratio:

$$\frac{\sigma_{\mathrm{B}}}{\sigma_{\mathrm{In}}} \sim \exp\!\left[\frac{(0.160 - 0.045)\ \mathrm{eV}}{0.05172\ \mathrm{eV}}\right] = \exp(2.224) = 9.24$$

$$\sigma(\mathbf{Si{:}B}) > \sigma(\mathbf{Si{:}In})$$

f) Germanium with 5 ppm of As or Sb

$$E_d(\mathrm{Sb}) = 0.0096\ \mathrm{eV} < E_d(\mathrm{As}) = 0.0127\ \mathrm{eV}$$

$$\frac{\sigma_{\mathrm{Sb}}}{\sigma_{\mathrm{As}}} \sim \exp\left[\frac{(0.0127 - 0.0096)\ \mathrm{eV}}{0.05172\ \mathrm{eV}}\right] = \exp(0.0599) = 1.062$$

$$\sigma(\mathbf{Ge{:}Sb}) > \sigma(\mathbf{Ge{:}As})$$

Result table

Item	Ranking, highest conductivity first	Criterion
a	InSb > GaAs > AlP	smaller E_g (larger atoms, more covalent)
b	InSb > CdTe	smaller E_g (III-V less ionic than II-VI)
c	Ge:Sb > Ge:P	smaller E_d
d	Ge:P > Si:P	smaller E_d and larger μ_e
e	Si:B > Si:In	smaller E_a
f	Ge:Sb > Ge:As	smaller E_d

RESULT & CHECKS

Checks

- **Dimensionless exponents.** Each exponent is an energy in eV divided by $2k_BT$ in eV, so it has no units; $[k_BT] = \mathrm{eV\,K^{-1}} \cdot \mathrm{K} = \mathrm{eV}$.
- **Arrows of Slide 40.** Donors: Sb (largest ion) above P, in both hosts, and Ge above Si; acceptors: B (smallest ion) above In. All agree with the ranking found.

- **Scale of the indicative ratios.** The extrinsic ratios in c, d and f are of the order of 1 to 6 because the levels are all of the order of $k_B T$ or smaller and the dopants are almost fully ionised; the pair e differs by 9 because the 0.160 eV of In is several times $k_B T$ (0.02586 eV), and In remains largely un-ionised.
- **The intrinsic ratios are many orders of magnitude** because the gaps are between 3 and 47 times $2k_B T$ (exponents of step a); they are indicative only, since the mobility and the prefactor also change from compound to compound (the mobility of InSb is 7.7 m²/(V·s), the largest on Slide 33, which strengthens the ranking).

Interpretation

Two rules cover the sheet: for intrinsic material the smaller gap wins, and the gap decreases with the size of the atoms and with the covalent character; for extrinsic material at the same concentration the dopant whose level is nearer to its band wins, with the host's mobility as a second factor. The values labelled "extension" (AlP, CdTe) affect only how far the intrinsic ratios go, not the order, which follows from the trend of Slides 31-32.

Check against the official solution

Item	Official	Computed	Comment
a to f	not printed on the sheet	table of the result	checked against the levels of Slide 40 and the gaps of Slide 33; the criterion is the one asked for on Slides 34 and 41